

A Data-Driven Examination of Global Emissions and Renewable Energy Adoption (2000–2021)





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Declaration

The information presented in this report is a unique creation by me and has not been presented, even partially, for other academic purposes. In addition, there are full legal references to all sources of information, data, and literature used in this report. I also confirm that I did this work without any outside assistance and by utilizing only legitimate sources.

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Abstract

This study presents a comprehensive data-driven analysis of global Greenhouse Gas (GHG) emissions and renewable energy adoption from 2000 to 2021. Two international datasets were cleaned, transformed, and standardized using SQL Server, where redundant metadata columns were removed, missing values were inspected, data types corrected, and analytical views created for integration into Power BI.

The development of a unified view on SQL made it possible to directly compare greenhouse gas emissions (GHGs) with renewable energy status (RE) indicators within any given country over multiple years. This was accomplished through the use of Power BI to create an interactive 3-page dashboard that utilizes advanced DAX functions for calculating averages and tracking instantaneous growth over the previous year (YOY) from 2015 to today, and creating composite scorecards of sustainability for each country based on their RE advancement and (GHG) emission levels. The graphics allow users to view trends across years, identify regional concentrations of GHG emissions and RE advancements, and make comparisons between countries based on their GHGs and RE advancements.

Analysis reveals large amounts of variability between some countries that are leading in RE and substantial contributors to GHGs, as well as the presence of an inverse relationship between GHG emissions and the adoption of RE, thus confirming the effectiveness of the clean energy initiatives. The results of this study illustrate how Power BI and SQL can be combined to provide high probability analytical outputs to monitor environmental issues and evaluate clean energy policy effectiveness. Limitations of this study include the lack of a complete data set for all three years researched, (no-adjusted per 1,000 per carbon emission standards) and thus no comparison across countries or expected GHGs from countries per capita. However, the study's methodology lays the groundwork for future improvements such as machine learning and predictive analysis.

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1. Introduction

Environmental sustainability has evolved into a significant global challenge as nations are pressured to lower Greenhouse Gas (GHG) emissions while rapidly transitioning to the use of renewable energy. In this scenario, monitoring the GHG emission levels and Renewable Energy Consumption on the basis of evidence-driven data will be key tools for Policymakers, Researchers, and International Agencies such as the United Nations (UN) and the International Energy Agency (IEA). In this paper, the use of SQL Server for data cleaning and Power BI for interactive visualisation of two worldwide datasets covering the period from 2000 to 2021 for GHG Emissions and Renewable Energy Consumption is explained. Additionally, a comprehensive analytical dashboard has been created utilizing advanced DAX metrics to allow for long-term global trend tracking, comparative country analysis, and the determination of the overall sustainability performance of the various countries according to the information provided by the output of the SQL Data Pre-processing in conjunction with the Power BI Model.

The final output is to consist of a three-page dashboard with detailed information relating to emissions, the progress made by countries in the use of renewable energy, and overall sustainability performance.

1.1 Objectives.

The following are the main objectives of this project:

- To view long-term global trends in emissions (2000 - 2021) by creating a visual representation of the changes in GHG emissions of all countries and regions.
- To provide the ability to view the level of adoption of renewables and the progress of each of the world's countries in achieving a higher percentage of renewable energy used as opposed to other more traditional methods.
- To compare each country's emissions with their level of adoption and use of renewables, allowing for the identification of the correlation of increased use of renewable energy to decreased emissions.
- To provide the user with interactive visualization tools such as filters, drilling down, etc., to allow the user to gain a deeper understanding of country-specific behaviours.
- To help generate insight into the data collected regarding energy and emissions to help support the academic and policy-making communities in the generation of knowledge regarding global environmental sustainability.

1.2 Purpose

This dashboard is designed to provide an interactive and comprehensive overview of worldwide emissions and renewable energy adoption spanning from 2000 - 2021. It assists users in identifying trends, making comparisons between countries, and understanding how the growth in renewable energy usage can help contribute to emissions reductions. Overall, the dashboard provides increased knowledge, assists academic researchers, and aids in the decision-making process for the environment at a global level.

2. Dataset Overview.

The researcher has selected two UN SDG Global Database datasets to conduct an analysis of both SDG 7 - Sustainable Energy for All, and SDG 13 - Climate Action. The purpose of this analysis is to investigate the temporal relationship between global adoption rates of renewable energy technologies and greenhouse gas emissions. Each dataset has comprehensive coverage of many countries over a lengthy time period (2000 to 2021) making them appropriate for conducting a time-series analysis using Power BI. This study used the following two datasets:

2.1 Greenhouse Gas (GHG) Emissions Dataset.

- **Dataset Description.**

The dataset provides information about the share of renewables in total energy consumption in each country. Among the sources of renewable energy are; wind, solar, hydropower, geothermal, and bioenergy. This data is often used to measure how well a country has been able to move toward cleaner, more sustainable energy sources.

- **Data Source.**

Source: UN SDG Global Database

Indicator Code: EG.FEC.RNEW.ZS

SDG Target: SDG 7 – Affordable and Clean Energy

Time Period: 2000–2021

Format: Excel (converted to SQL table)

- **Key Variables.**

Column	Description
Series Code	Unique SDG indicator code (e.g. EN.ATM.GHGT.AIP)
Series Description	Description of the emissions indicator
Country Code	UN numeric or alphanumeric country code
Country	Name of the reporting country

Year	Reporting year (2000-2021)
Value	Total GHG emissions for that year (CO ₂ -eq)

Table 1. key variables of Greenhouse Gas (GHG) Emissions Dataset.

2.2 Renewable Energy Consumption Dataset

- **Dataset Description.**

The dataset consists of the information regarding percentage share of Renewable Energy in Total Final Energy Consumption for each country around the world. Types of Renewable Energy Sources include Wind/Solar/Hydro/Geyser and Biomass. This data set has a great deal of use in monitoring countries' national advancement on their capacity to transition to a more sustainable and less polluting energy system.

- **Data Source.**

Source: UN SDG Global Database

Indicator Code: EG.FEC.RNEW.ZS

SDG Target: SDG 7 – Affordable and Clean Energy

Time Period: 2000–2021

Format: Excel (converted to SQL table)

- **Key Variables.**

Column	Description
Series Code	Indicator code for renewable energy (% of total consumption)
Series Description	Describes the renewable energy indicator
Country Code	UN country code matching the emissions dataset
Country	Reporting country
Year	Reporting year (2000–2021)
Value	Share of renewable energy consumption (percentage)

Table 2. Key variables of Renewable Energy Consumption Dataset

3. Methodology

3.1 Download and Import Data

Visit the **UN SDG Data Platform** or **UN Data Portal** (data.un.org). Use the link below.

Link; <https://unstats.un.org/sdgs/dataportal>

Go through the above link and Explore datasets covering one or more SDGs. For this dashboard we used two different SDGs. The website can be seen as follows and go to the search bar and find the dataset as you need.

In here I recommend Clean Energy + CO₂ Emissions (SDG 7 + SDG 13) two combined datasets. It is best for Environmental sustainability, energy transition, climate impact analysis. Following 2 datasets are we used to create this dashboard.

- SDG 7.2.1 – Renewable energy share in total final energy consumption (%).
- SDG 13 – CO₂ emissions (per capita).



Figure 1. UN SDG Data Platform.

The downloaded two datasets are as follows. Before stored in SQL server database download the above two datasets in CSV or Excel format. And before importing to SQL database get clear ideas about datasets.

- Renewable energy share in total final energy consumption dataset.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
Goal	Target	Indicator	SeriesCode	SeriesDesc	GeoAreaC	GeoAreaN	TimePeriod	Value	Time_Det	TimeCover	UpperBou	LowerBou	BasePeriod	Source	GeoInfoUr	FootNote	Nature	Reporting	Units
7	7.2	7.2.1	EG_FEC_R	Renewable	4	Afghanistan	2000	44.99	2000					Energy Balances, UN Statistics D	E		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	4	Afghanistan	2001	45.6	2001					Energy Balances, UN Statistics D	E		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	4	Afghanistan	2002	37.83	2002					Energy Balances, UN Statistics D	E		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	4	Afghanistan	2003	36.66	2003					Energy Balances, UN Statistics D	E		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	4	Afghanistan	2004	44.24	2004					Energy Balances, UN Statistics D	E		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	4	Afghanistan	2005	33.88	2005					Energy Balances, UN Statistics D	E		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	4	Afghanistan	2006	31.89	2006					Energy Balances, UN Statistics D	E		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	4	Afghanistan	2007	28.78	2007					Energy Balances, UN Statistics D	E		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	4	Afghanistan	2008	21.17	2008					Energy Balances, UN Statistics D	E		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	4	Afghanistan	2009	16.53	2009					Energy Balances, UN Statistics D	E		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	4	Afghanistan	2010	15.15	2010					Energy Balances, UN Statistics D	E		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	4	Afghanistan	2011	12.61	2011					Energy Balances, UN Statistics D	E		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	4	Afghanistan	2012	15.36	2012					Energy Balances, UN Statistics D	E		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	4	Afghanistan	2013	16.89	2013					Energy Balances, UN Statistics D	E		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	4	Afghanistan	2014	19.06	2014					Energy Balances, UN Statistics D	E		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	4	Afghanistan	2015	17.69	2015					Energy Balances, UN Statistics D	E		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	4	Afghanistan	2016	20.18	2016					Energy Balances, UN Statistics D	E		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	4	Afghanistan	2017	19.48	2017					Energy Balances, UN Statistics D	E		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	4	Afghanistan	2018	18.28	2018					Energy Balances, UN Statistics D	E		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	4	Afghanistan	2019	18.87	2019					Energy Balances, UN Statistics D	E		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	4	Afghanistan	2020	18.21	2020					Energy Balances, UN Statistics D	E		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	4	Afghanistan	2021	19.98	2021					Energy Balances, UN Statistics D	E		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	4	Afghanistan	2022	20.02	2022					Energy Balances, UN Statistics D	E		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	2	Africa	2000	52.11	2000					IEA (2024), World Energy Balance	N		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	2	Africa	2001	51.54	2001					IEA (2024), World Energy Balance	N		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	2	Africa	2002	51.32	2002					IEA (2024), World Energy Balance	N		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	2	Africa	2003	50.7	2003					IEA (2024), World Energy Balance	N		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	2	Africa	2004	50.47	2004					IEA (2024), World Energy Balance	N		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	2	Africa	2005	49.44	2005					IEA (2024), World Energy Balance	N		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	2	Africa	2006	49.29	2006					IEA (2024), World Energy Balance	N		G	PERCENT	
7	7.2	7.2.1	EG_FEC_R	Renewable	2	Africa	2007	48	2007					IEA (2024), World Energy Balance	N		G	PERCENT	

Figure 2. Renewable energy share in total final energy consumption dataset.

- CO₂ emissions dataset.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
Goal	Target	Indicator	SeriesCode	SeriesDesc	GeoAreaC	GeoAreaN	TimePeriod	Value	Time_Det	TimeCover	UpperBou	LowerBou	BasePeriod	Source	GeoInfoUr	FootNote	Nature	Reporting	Units
13	13.2	13.2.2	EN_ATM_(Total gree		36	Australia	2000	489.5287	2000					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		36	Australia	2001	497.352	2001					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		36	Australia	2002	501.0008	2002					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		36	Australia	2003	501.129	2003					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		36	Australia	2004	518.4084	2004					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		36	Australia	2005	524.8119	2005					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		36	Australia	2006	529.2406	2006					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		36	Australia	2007	535.8076	2007					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		36	Australia	2008	538.5907	2008					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		36	Australia	2009	541.4758	2009					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		36	Australia	2010	536.8937	2010					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		36	Australia	2011	538.6639	2011					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		36	Australia	2012	541.8997	2012					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		36	Australia	2013	532.2673	2013					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		36	Australia	2014	526.7118	2014					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		36	Australia	2015	534.936	2015					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		36	Australia	2016	543.9768	2016					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		36	Australia	2017	550.8749	2017					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		36	Australia	2018	552.484	2018					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		36	Australia	2019	546.6066	2019					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		36	Australia	2020	528.1495	2020					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		36	Australia	2021	528.1495	2021					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		40	Austria	2000	80.61936	2000					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		40	Austria	2001	84.51405	2001					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		40	Austria	2002	86.26754	2002					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		40	Austria	2003	91.6363	2003					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		40	Austria	2004	91.4279	2004					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		40	Austria	2005	92.58858	2005					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		40	Austria	2006	90.15867	2006					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		40	Austria	2007	87.37795	2007					UNFCCC		Party inclu C	G	MT_CO2E	
13	13.2	13.2.2	EN_ATM_(Total gree		40	Austria	2008	86.77082	2008					UNFCCC		Party inclu C	G	MT_CO2E	

Figure 3. CO₂ emissions dataset.

3.2 Database Design

- **Create a new database.**
 - a. Open SSMS and connect to SQL server instance.
 - b. Create a new database called “UN_RenewClimate_DB”.
 - c. Open New Query > CREATE DATABASE “UN_RenewClimate_DB” > Execute.

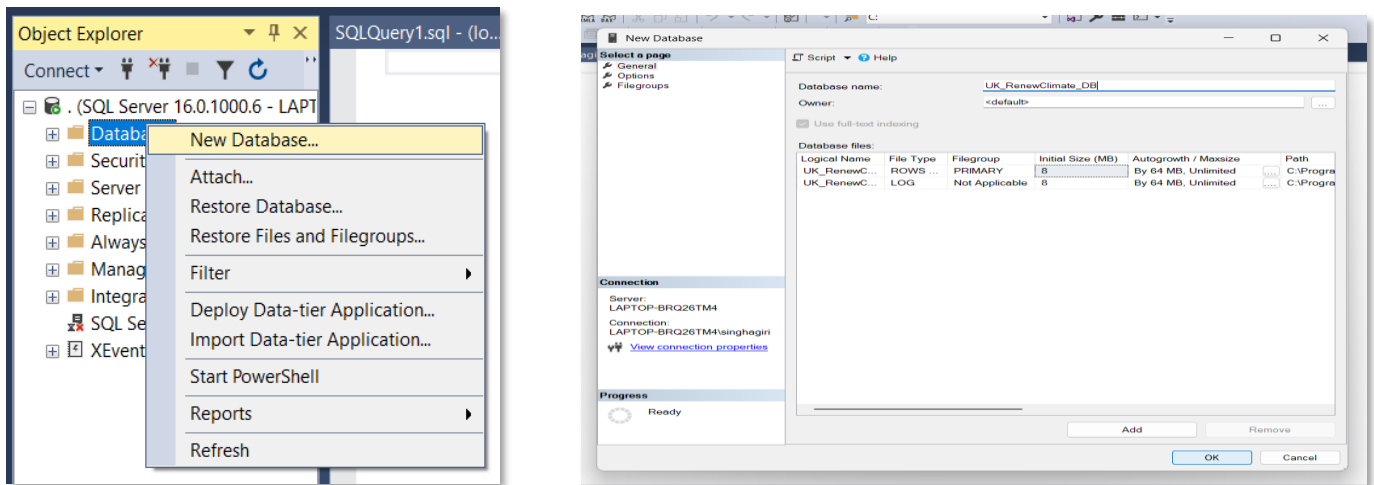


Figure 4. Create a new database

- **Import Dataset**

Import the dataset CSV file to the SQL server database “UN_RenewClimate_DB”.

- a. Right-click the “UN_RenewClimate_DB” database.
- b. Select Tasks → Import flat file.

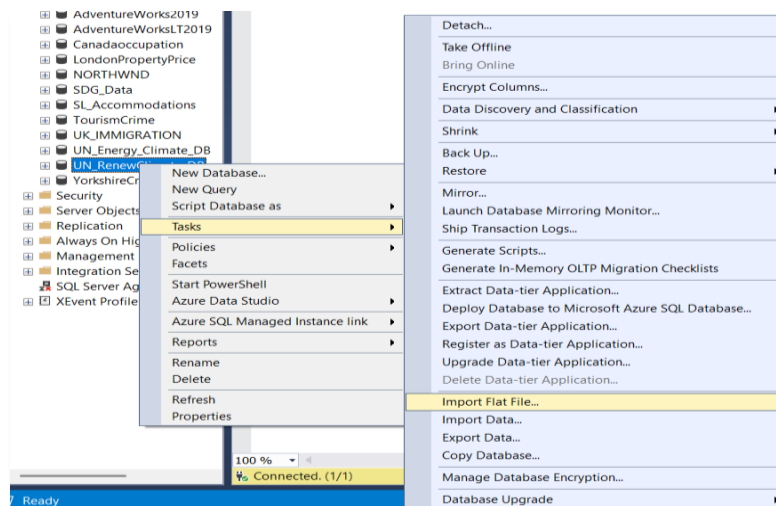


Figure 5. Import Dataset

c. Then click 'NEXT'

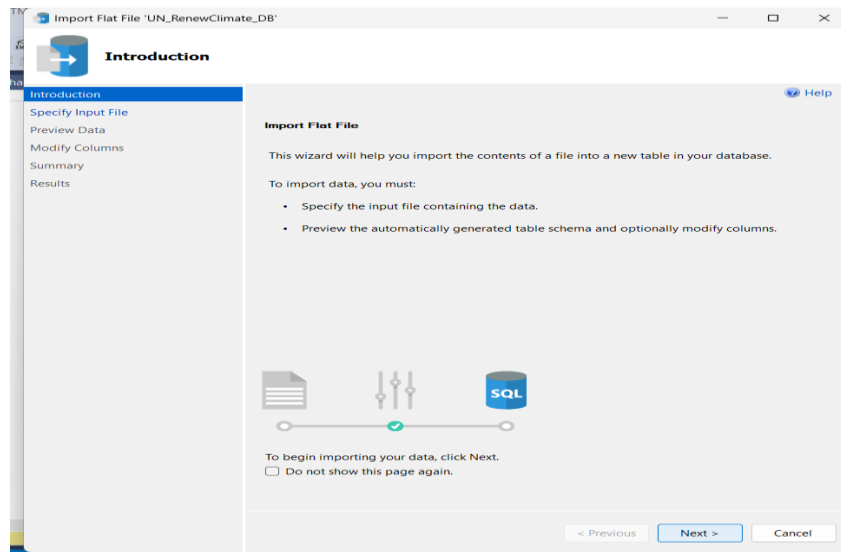


Figure 6. Import flat file

d. Then click on Browse.

Click on browse and select your dataset that you needed and then go next.

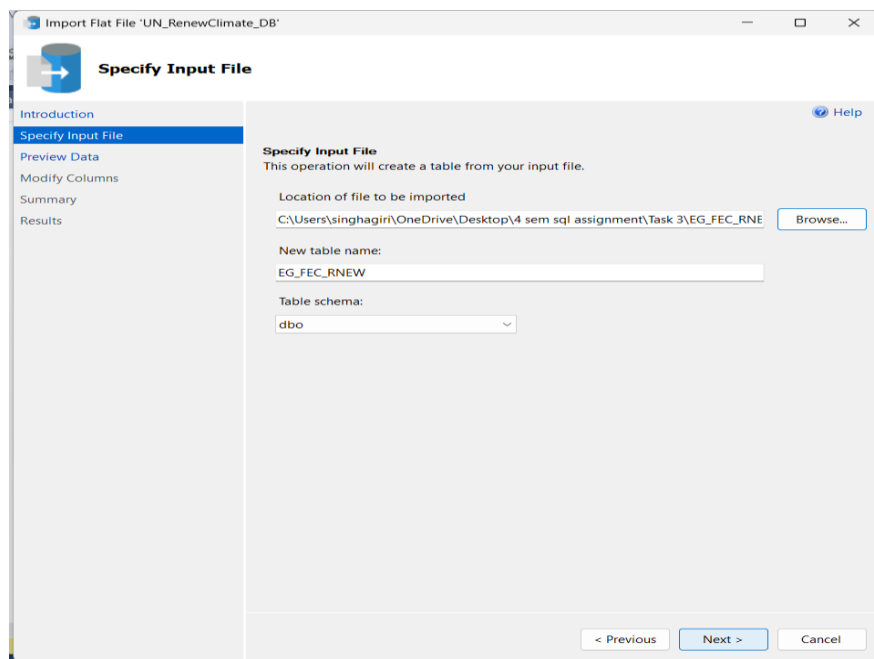


Figure 7. Specify input file

e. Click 'NEXT'

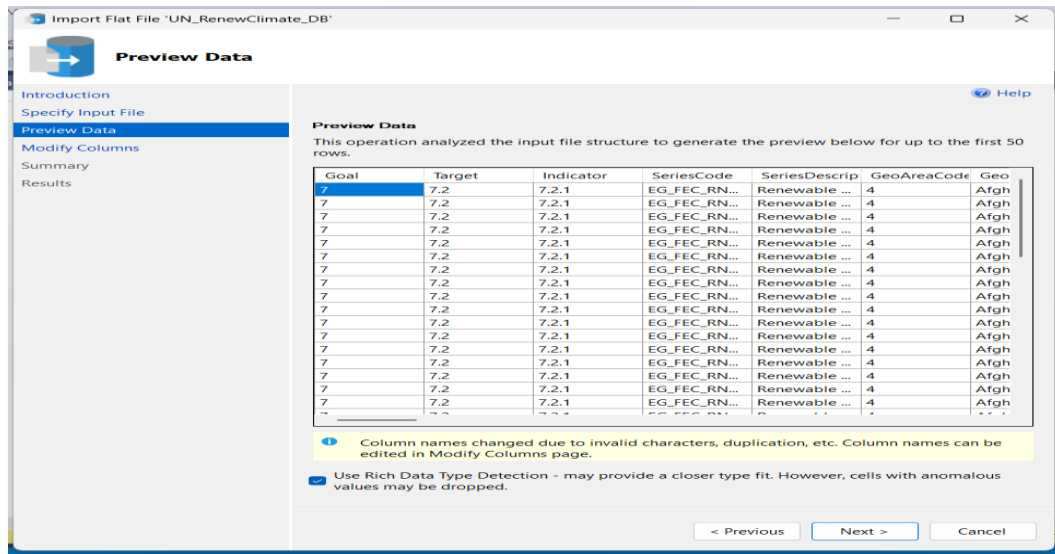


Figure 8. Preview data

- f. modify columns.

change the data type and then Allow nulls for all data. Then click 'NEXT'.

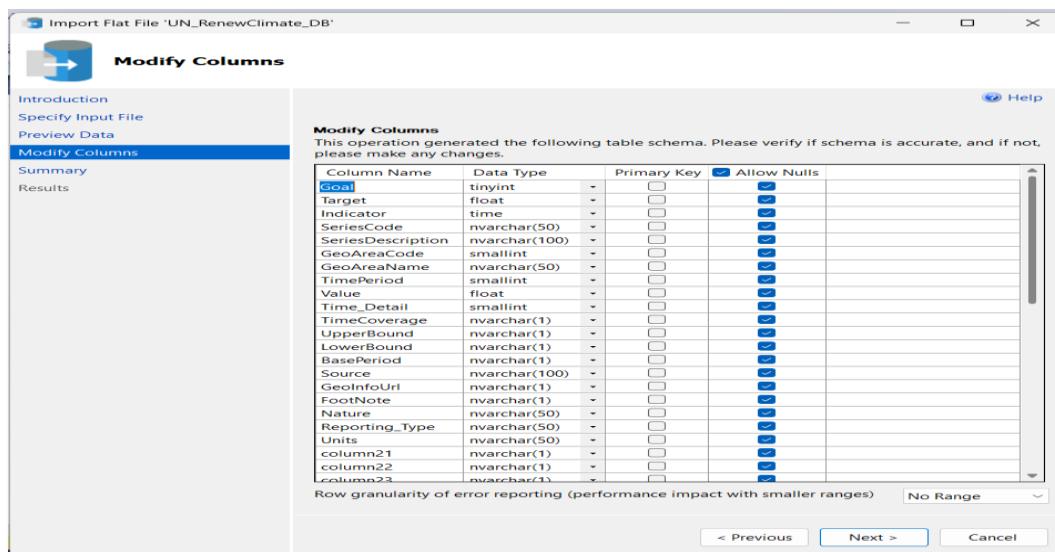


Figure 9. Modify columns

g. Summary and Result.

Click finish and import the datasets to SQL database successfully. If there is an error the dataset doesn't import then click previous and correct the error.

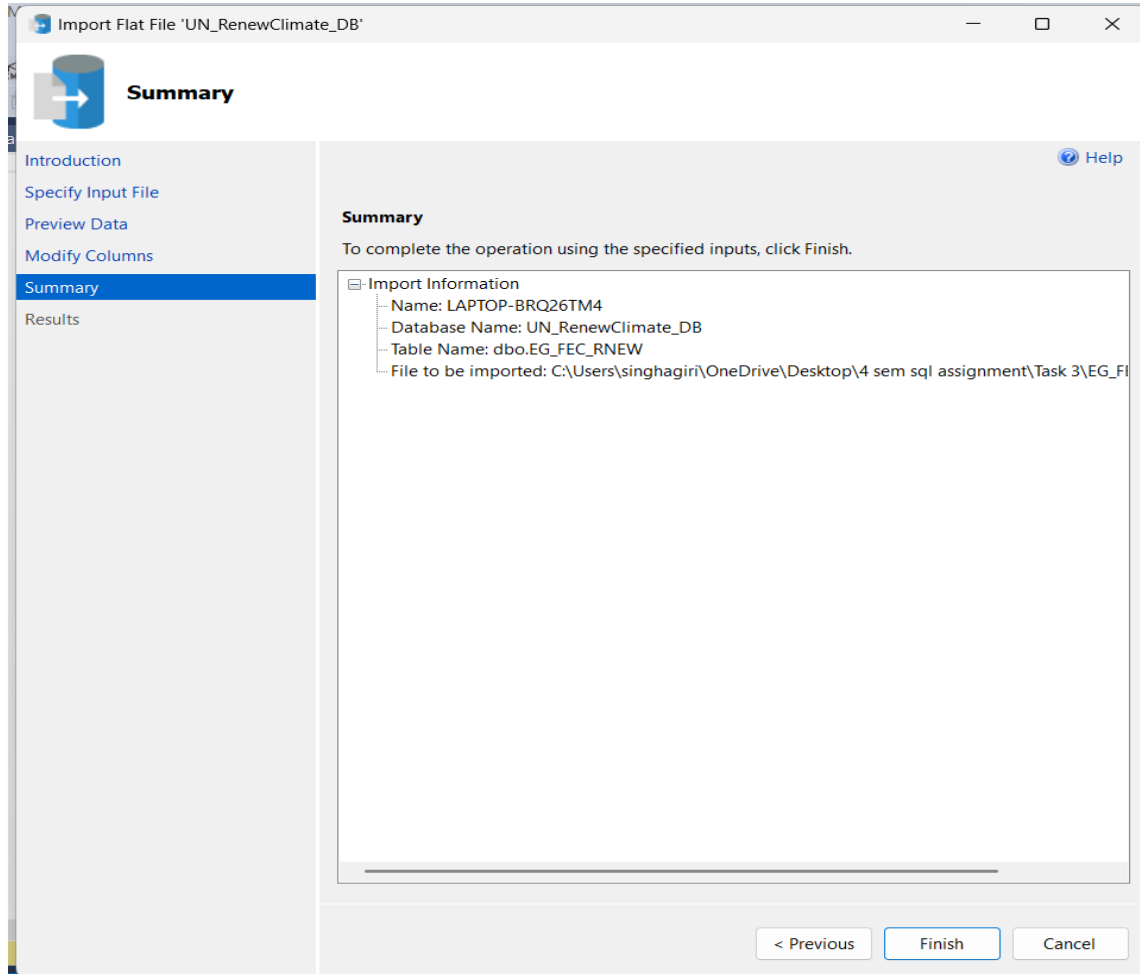


Figure 10. Summary

Repeat these all steps for **Renewable energy share in total final energy consumption dataset** and **CO₂ emissions dataset**.

3.3 SQL Cleaning.

Open SQL Query and clean the import datasets by using following codes.

- **Rename Tables.**

```
--Step 1 - Rename tables  
EXEC sp_rename '[dbo].[EN_ATM_GHGT_AIP]', 'GHG_Emissions';  
EXEC sp_rename '[dbo].[EG_FEC_RNEW]', 'Renewable_Energy';
```

Figure 11. Rename Tables

The file names associated with the data set files that are downloaded from the UN SDG Global Database consist, for the most part, of long or technical names representing the SDG Indicator Codes. These lengthy and/or technical names do not provide easy access for SQL Development and Dashboard Analytics Users. The process of renaming these tables will provide users with clearer and more understanding Table Identifiers as well as increasing the user ability to read the tables easily which ultimately improves the likelihood of an error in writing SQL Queries. Therefore, effective renaming of the tables will enhance the clarity, maintainability, lessen errors in SQL Queries, and provide a standardised naming system for the tables.

- **Remove Unwanted Columns.**

```
--Step 2 - Remove unwanted columns  
--For GHG_Emissions  
ALTER TABLE GHG_Emissions DROP COLUMN [Goal];  
ALTER TABLE GHG_Emissions DROP COLUMN [Target];  
ALTER TABLE GHG_Emissions DROP COLUMN [Indicator];  
ALTER TABLE GHG_Emissions DROP COLUMN [Time_Detail];  
ALTER TABLE GHG_Emissions DROP COLUMN [TimeCoverage];  
ALTER TABLE GHG_Emissions DROP COLUMN [UpperBound];  
ALTER TABLE GHG_Emissions DROP COLUMN [LowerBound];  
ALTER TABLE GHG_Emissions DROP COLUMN [BasePeriod];  
ALTER TABLE GHG_Emissions DROP COLUMN [Source];  
ALTER TABLE GHG_Emissions DROP COLUMN [GeoInfoUrl];  
ALTER TABLE GHG_Emissions DROP COLUMN [FootNote];  
ALTER TABLE GHG_Emissions DROP COLUMN [Nature];  
ALTER TABLE GHG_Emissions DROP COLUMN [Reporting_Type];  
ALTER TABLE GHG_Emissions DROP COLUMN [Units];
```

Figure 12. Remove Unwanted Columns

Once the tables had been renamed, the following task was to prepare the dataset to be analysed by eliminating any redundant or irrelevant columns that do not add to the analytical objectives as outlined in the dashboard. The raw dataset that is imported from the UN Sustainable Development Goals (SDG) Global Database comprises various types of metadata, including indicator codes, reporting notes, data quality descriptions, and geographical information URL's. Even though some of these metadata fields may have documentation benefits, they are not appropriate for numerical analysis, trend comparison, or any type of visual representation within Power BI. Therefore, to streamline the dataset and keep only the most pertinent variables, the above-mentioned SQL statements were executed to remove the unrelated columns.

- **Standardize Column names.**

```
--Step 3 - Standardize column names
--For GHG_Emissions
EXEC sp_rename 'GHG_Emissions.[SeriesCode]', 'Series_Code';
EXEC sp_rename 'GHG_Emissions.[SeriesDescription]', 'series_Description';
EXEC sp_rename 'GHG_Emissions.[GeoAreaCode]', 'Country_Code';
EXEC sp_rename 'GHG_Emissions.[GeoAreaName]', 'Country';
EXEC sp_rename 'GHG_Emissions.[TimePeriod]', 'Year';

--For Renewable_Energy
EXEC sp_rename 'Renewable_Energy.[SeriesCode]', 'Series_Code';
EXEC sp_rename 'Renewable_Energy.[SeriesDescription]', 'Series_Description';
EXEC sp_rename 'Renewable_Energy.[GeoAreaCode]', 'Country_Code';
EXEC sp_rename 'Renewable_Energy.[GeoAreaName]', 'Country';
EXEC sp_rename 'Renewable_Energy.[TimePeriod]', 'Year';
```

Figure 13. Standardize Column names

The initial step of cleaning the data involved deleting every one of the redundant columns containing excessive amounts of metadata. The next step was to standardize all of the column headings across both data sources. The column headings associated with the UN Sustainable Development Goals (SDG) Database were characterized by long, often very long, technical or strange naming conventions, which may have been inconsistent across multiple data files. Additionally, the column headings are often not very obvious when attempting to create an SQL query or produce a report using Power Bi. For the purpose of establishing consistencies and better overall readability, we provided more descriptive names for all of the key columns from both datasets, thus simplifying the process of joining, comparing, and maintaining the two datasets when working with Power Bi.

- **Check data types.**

```
--Step 4 - Check data types
--For GHG_Emissions
SELECT
    COLUMN_NAME,
    DATA_TYPE,
    CHARACTER_MAXIMUM_LENGTH
FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_NAME = 'GHG_Emissions';

--For Renewable_Energy
SELECT
    COLUMN_NAME,
    DATA_TYPE,
    CHARACTER_MAXIMUM_LENGTH
FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_NAME = 'Renewable_Energy';
```

Figure 14. Check data types

After cleansing and standardizing the dataset, the following phase is to check the “Data Type” columns from each of the two tables. During the importation of datasets (Excel or CSV) into SQL Server, SQL Server frequently assigns an inaccurate or dissimilar “Data Type” for each column. Therefore, the first step is to validate the Data Type of each column to ensure its applicable use prior to commencing further cleaning and analysis. Furthermore, by reviewing the Data Type, this provides three different benefits: 1) identification of inaccurate Data Type, 2) compatibility with Power BI, 3) preservation of data integrity with the use of appropriate “Data Type”. Without proper Data Types, you sacrifice several critical features of Power BI, including; Time-Series Line Graphs, Country-Level Comparisons, Trend Analysis, KPI Measure, and Year Over Year Measure.

- **Apply the correct data types.**

```
--Step 5 - Apply the correct data types
--For GHG_Emissions
ALTER TABLE GHG_Emissions ALTER COLUMN Series_Code VARCHAR(50);
ALTER TABLE GHG_Emissions ALTER COLUMN Series_Description VARCHAR(255);

ALTER TABLE GHG_Emissions ALTER COLUMN Country_Code VARCHAR(10);
ALTER TABLE GHG_Emissions ALTER COLUMN Country VARCHAR(100);

ALTER TABLE GHG_Emissions ALTER COLUMN Year INT;
ALTER TABLE GHG_Emissions ALTER COLUMN Value FLOAT;

--For Renewable_Energy
SELECT [Value]
FROM Renewable_Energy
WHERE TRY_CONVERT(float, [value]) IS NULL
AND [Value] IS NOT NULL;

ALTER TABLE Renewable_Energy ALTER COLUMN Series_Code VARCHAR(50);
ALTER TABLE Renewable_Energy ALTER COLUMN Series_Description VARCHAR(255);

ALTER TABLE Renewable_Energy ALTER COLUMN Country_Code VARCHAR(10);
ALTER TABLE Renewable_Energy ALTER COLUMN Country VARCHAR(100);

ALTER TABLE Renewable_Energy ALTER COLUMN Year INT;
ALTER TABLE Renewable_Energy ALTER COLUMN Value float;
```

After completing a detailed inspection of the column structure, our next own task was to ensure that all columns contained the correct data type during the importation into SQL Server because SQL Server will select a default column data type when it imports files, many times causing errors or incorrect data type formatting; if you do not have the proper data types in your columns, you can experience errors in calculations, invalid visualizations within Power BI, relationships that do not connect between tables, and performance problems. So the selection of the correct data type prior to your analysis is an essential part of getting your dataset ready for use in analysis.

- **Find NULL Values**

```
--Find rows containing ANY NULL values
SELECT *
FROM GHG_Emissions
WHERE
    Series_Code IS NULL OR
    Series_Description IS NULL OR
    Country_Code IS NULL OR
    Country IS NULL OR
    Year IS NULL OR
    Value IS NULL;

Select *
From GHG_Emissions

SELECT *
FROM Renewable_Energy
WHERE
    series_Code IS NULL OR
    Series_Description IS NULL OR
    Country_Code IS NULL OR
    Country IS NULL OR
    Year IS NULL OR
    Value IS NULL;

Select *
From Renewable_Energy
```

Figure 16. Find NULL Values

After correcting the data types, the next step was to check the dataset for missing or null values. Null values can cause serious issues during data analysis, such as: Incorrect averages and totals, Breaks in time-series charts, Power BI errors (e.g., “Cannot aggregate null values”). Therefore, it is essential to detect any null values before loading the data into Power BI. This step is important to Ensures the dataset is complete, Avoids issues in Power BI, Prepares for clean data modelling. By detecting null values early, the dataset becomes much easier to model, and the final dashboard becomes more accurate and visually consistent.

- **Check Duplicate Rows**

```
--Check Duplicate Rows
SELECT
    Series_Code, Series_Description, Country_Code, Country, Year, Value,
    COUNT(*) AS Duplicate_Count
FROM GHG_Emissions
GROUP BY
    Series_Code, Series_Description, Country_Code, Country, Year, Value
HAVING COUNT(*) > 1;

SELECT
    Series_Code, Series_Description, Country_Code, Country, Year, Value,
    COUNT(*) AS Duplicate_Count
FROM Renewable_Energy
GROUP BY
    Series_Code, Series_Description, Country_Code, Country, Year, Value
HAVING COUNT(*) > 1;

Select *
From GHG_Emissions

Select *
From Renewable_Energy
```

Figure 17. Check Duplicate Rows

The SQL queries searched for any duplicate records that had not been removed from the data set after cleaning. According to the organization's definition, a duplicate record is one where each of the six major fields (Series_Code, Series_Description, Country_Code, Country, Year, and Value) contain exactly identical information. Therefore, duplicate checking was performed using the SQL functions GROUP BY, COUNT(*) , and HAVING COUNT(*) > 1. By performing this check, the organization was able to confirm that none of the records had the same information for all six of the fields used to define a duplicate record. This is an important part of the data cleansing process to ensure that the data used for building Power BI graphics is correct. There were no records which were marked as duplicate in either data set.

3.4 Create Aggregation Views

A SQL View provides a **clean, ready-to-use dataset** for Power BI. It standardizes: Column names, Data types, Removes NULL values in the Value field. Power BI can connect directly to the view instead of the raw table. This improves **performance, cleanliness, and reproducibility** of the dashboard.

- **Renewable Energy View**

```
--Create SQL Views for Power BI?
--View 1: vw_Renewable_Energy
CREATE VIEW vw_Renewable_Energy AS
SELECT
    Series_Code,
    Series_Description,
    Country_Code,
    Country,
    CAST(Year AS INT) AS Year,
    CAST(Value AS FLOAT) AS Renewable_Energy_Percent
FROM Renewable_Energy
WHERE Value IS NOT NULL;
```

Figure 18. Renewable Energy View

- **Green House Emissions View**

```
--View 2: vw_GHG_Emissions
CREATE VIEW vw_GHG_Emissions AS
SELECT
    Series_Code,
    Series_Description,
    Country_Code,
    Country,
    CAST(Year AS INT) AS Year,
    CAST(Value AS FLOAT) AS GHG_Emissions
FROM GHG_Emissions
WHERE Value IS NOT NULL;
```

Figure 19. Green House Emissions View

- **Combine View**

```
--Create a Combined View for Power BI
CREATE VIEW vw_RE_GHG_Combined AS
SELECT
    r.Country,
    r.Country_Code,
    r.Year,
    r.Renewable_Energy_Percent,
    g.GHG_Emissions
FROM vw_Renewable_Energy r
INNER JOIN vw_GHG_Emissions g
ON r.Country = g.Country
AND r.Year = g.Year;
```

Figure 20. Combine View

3.5 Power BI Dashboard design and implementation.

1. Open Power BI. This is Powe BI interface.

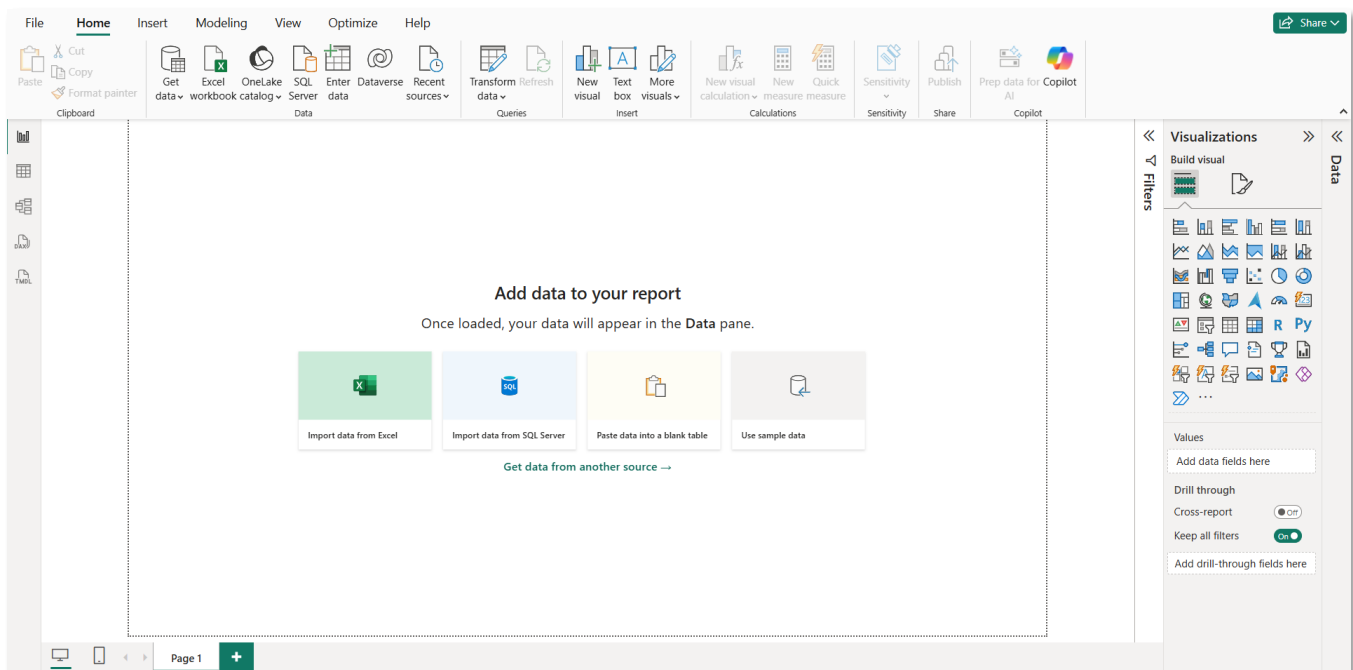


Figure 21. 1.Open Power BI

2. Import data into Power BI. Go to Get data and select SQL server.

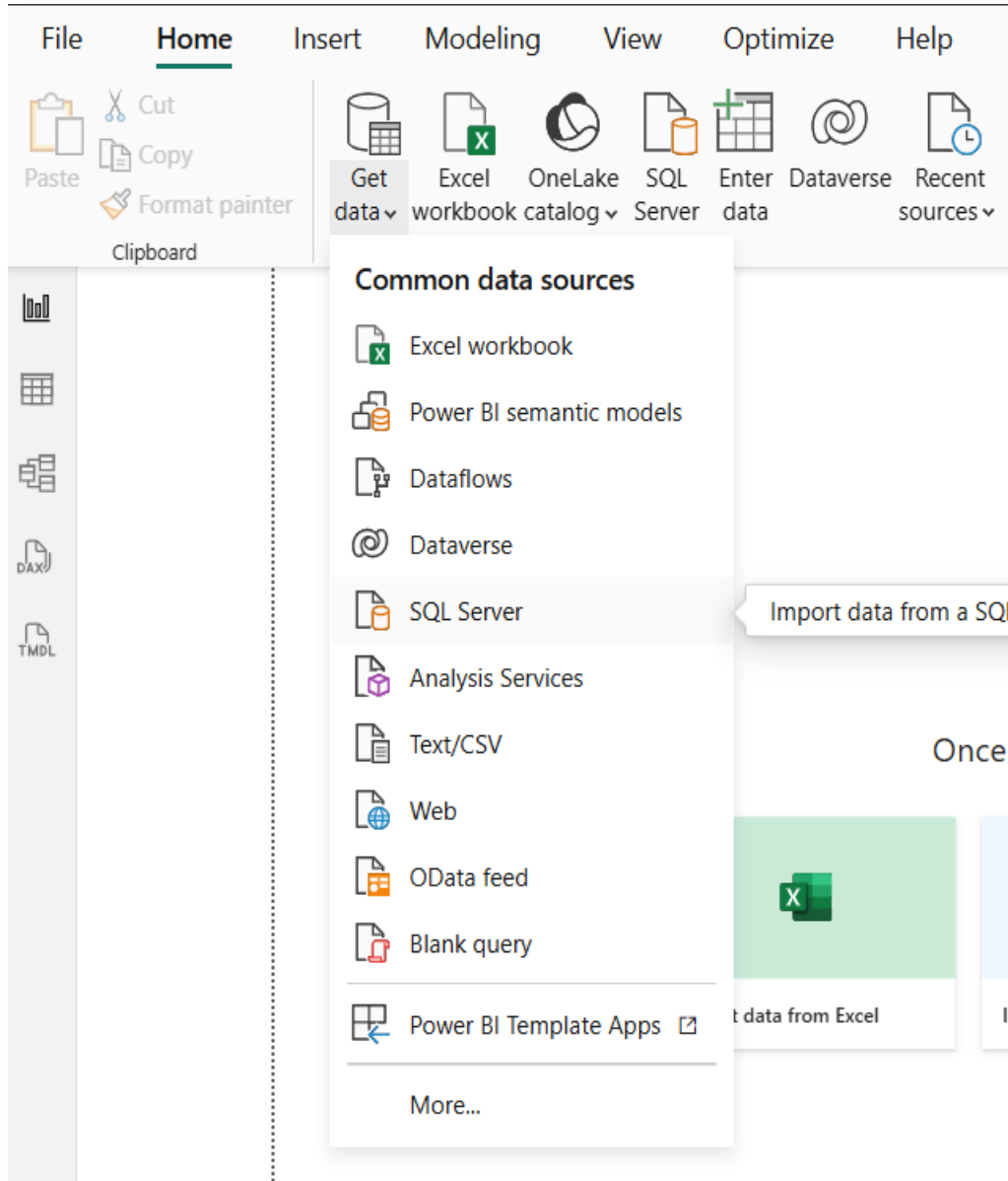


Figure 22.Import data into Power BI

3. From SQL Server Database Server connect to your server. Then select Import and click OK.

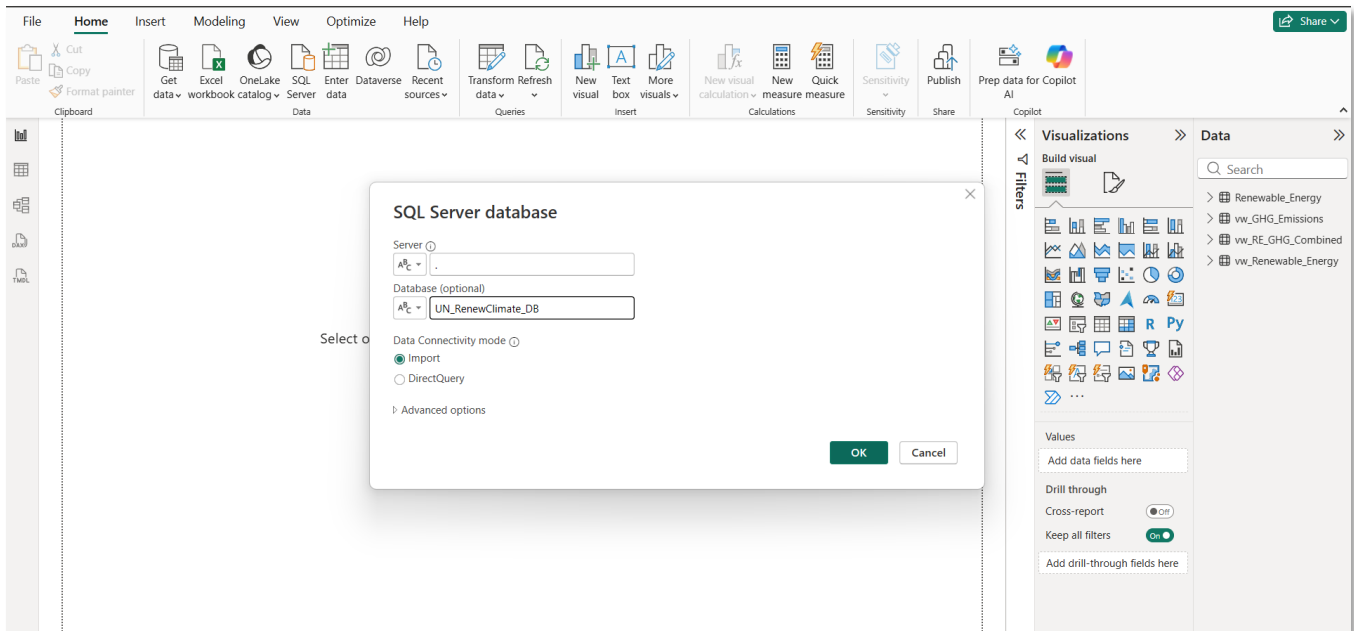


Figure 23. From SQL Server Database Server connect to your server

4. In Navigation Tab, select UN RenewClimate DB Database and Transform data.

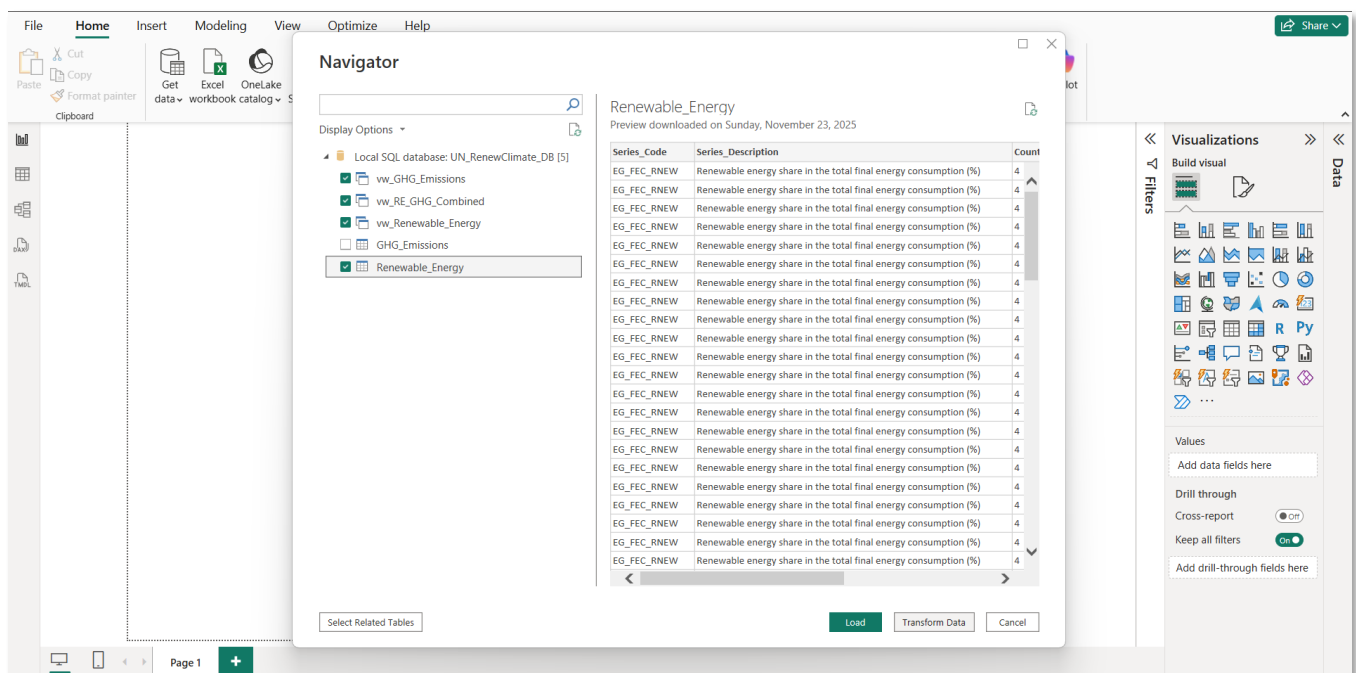


Figure 24. select UN RenewClimate_DB Database and Transform data.

5. After loading data Use the Power Query Editor to edit as you need. After that Close and apply. Then you can see Data View in right corner.

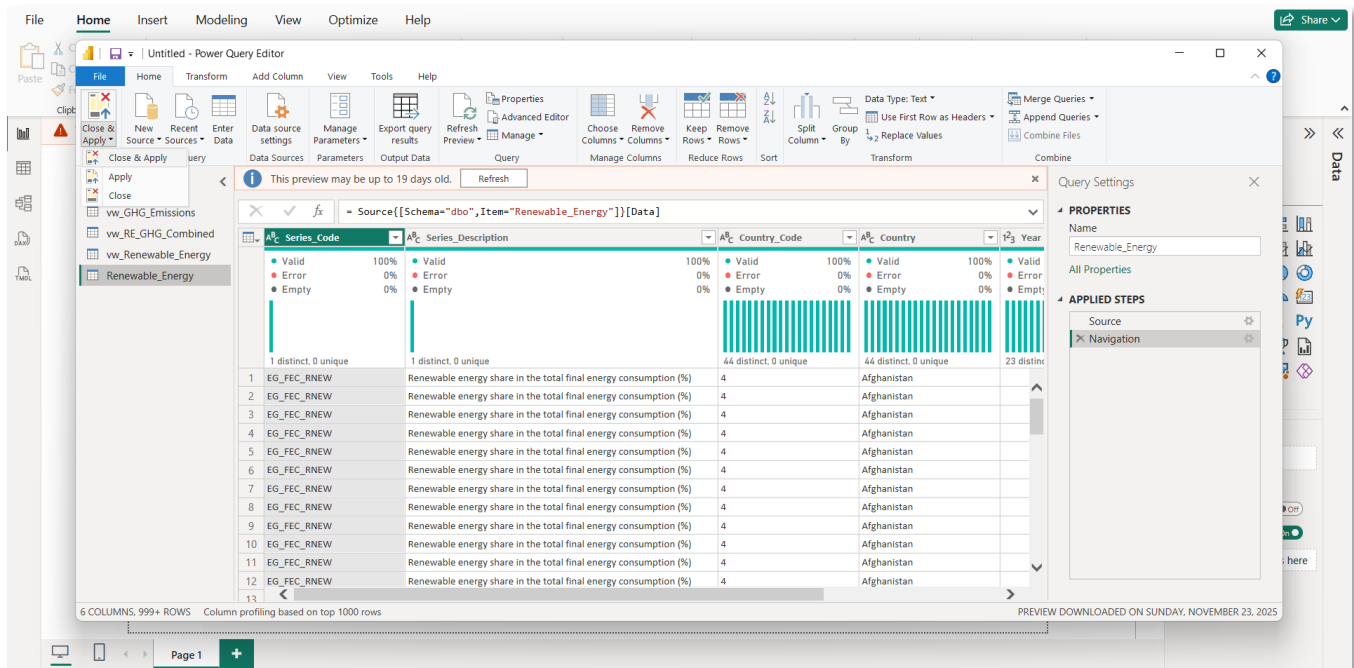


Figure 25. the Power Query Editor to edit

3.6 Build new DAX measurements and new column.

- **Create Dimensional Tables**

TABLE 1 — Dim_Country

This table is used to avoid repeating country names in both datasets and to link both tables through one shared dimension.

```
1 DimCountry =  
2 DISTINCT(  
3     UNION(  
4         SELECTCOLUMNS(GHG_Emissions, "Country", GHG_Emissions[Country], "Country_Code", GHG_Emissions[Country_Code]),  
5         SELECTCOLUMNS(Renewable_Energy, "Country", Renewable_Energy[Country], "Country_Code", Renewable_Energy[Country_Code])  
6     )  
7 )
```

Figure 26. Dim_Country

TABLE 2 — Dim_Indicator (Series Table)

Green House Gas Emission and Renewable Energy both have indicators (Series Code, Series Description). So, you need a unified indicator dimension.

```
1 DimIndicator =  
2 DISTINCT (  
3     UNION(  
4         SELECTCOLUMNS(GHG_Emissions, "Series_Code", GHG_Emissions[Series_Code],  
5         | | | | "Series_Description", GHG_Emissions[series_Description]),  
6         SELECTCOLUMNS(Renewable_Energy, "Series_Code", Renewable_Energy[Series_Code],  
7         | | | | "Series_Description", Renewable_Energy[Series_Description])  
8     )  
9 )
```

Figure 27. Dim_Indicator

TABLE 3 — Dim_Year

You need this for: YoY calculations, Rolling averages, Cumulative progress, Time intelligence (DATEADD, YTD).

```
1 DimYear =  
2 DISTINCT (  
3     UNION (  
4         SELECTCOLUMNS(GHG_Emissions, "Year", GHG_Emissions[Year]),  
5         SELECTCOLUMNS(Renewable_Energy, "Year", Renewable_Energy[Year])  
6     )  
7 )
```

Figure 28. Dim_Year

- **DAX Measurements.**

1. AVG Emission & AVG Renewable

This measure calculates the overall average emission & Renewable energy value in the dataset. It helps compare whether a country is performing above or below the global average.

```
Avg Emission = AVERAGE(GHG_Emissions[Value])
```

```
Avg Renewable = AVERAGE(Renewable_Energy[Value])
```

Figure 29. AVG Emission & AVG Renewable

2. Avg GHG Indicator & Avg Renewable Indicator

This measure tells us the average greenhouse gas emission level & Renew Energy level across all available countries and years in the dataset.

```
Avg GHG Indicator =  
AVERAGE ( 'GHG_Emissions'[Value] )
```

```
Avg Renewable Indicator =  
AVERAGE ( 'Renewable_Energy'[Value] )
```

Figure 30. Avg GHG Indicator & Avg Renewable Indicator

3. Renewable Target Progress

This measure shows the percentage of the renewable energy target achieved. For example, 0.75 means the country has achieved 75% of its target.

```
Progress vs Target (%) =  
DIVIDE([Avg Renewable Indicator], [Renewable Target])
```

Figure 31. Renewable Target Progress

4. Emission Rank & Renewable Rank

This measure ranks countries from highest to lowest based on their average greenhouse gas emissions & Renewable energy. A rank of 1 means the country has the highest emissions.

```
Emission Rank =  
RANKX(ALL(GHG_Emissions[Country]), [Avg Emission], , DESC)  
  
Renewable Rank =  
RANKX(ALL(DimCountry[Country]), [Avg Renewable Indicator], , DESC)
```

Figure 32. Emission Rank & Renewable Rank

5. Emission Since 2015 & Renewable Since 2015

This measure shows how much emissions & Renewable energy have increased or decreased since 2015. Interpretation:

- Positive value → emissions increased since 2015 (worse).
- Negative value → emissions decreased since 2015 (improvement).
- 0 → no change.

```
Emission Since 2015 =  
[Avg GHG Indicator] -  
CALCULATE([Avg GHG Indicator], DimYear[Year] = 2015)  
  
Renewable Since 2015 =  
[Avg Renewable Indicator] -  
CALCULATE([Avg Renewable Indicator], DimYear[Year] = 2015)
```

Figure 33. Emission Since 2015 & Renewable Since 2015

6. Emission YOY% & Renewable YOY%

This measure shows the percentage increase or decrease in emissions & renew energy compared to the previous year.

```
Emissions YoY % =  
VAR PrevYear =  
    CALCULATE(  
        [Total Emission],  
        GHG_Emissions[Year] = SELECTEDVALUE(GHG_Emissions[Year]) - 1  
    )  
RETURN  
    DIVIDE([Total Emission] - PrevYear, PrevYear)  
  
Renewable_YoY % =  
VAR PrevYear =  
    CALCULATE(  
        [Total Renewable],  
        Renewable_Energy[Year] = SELECTEDVALUE(Renewable_Energy[Year]) - 1  
    )  
RETURN  
    DIVIDE([Total Renewable] - PrevYear, PrevYear)
```

Figure 34. Emission YOY% & Renewable YOY%

7. Emission YOY Change & Renewable YOY Change

This measure shows how much emissions & renew energy have increased or decreased compared to the previous year.

Interpretation:

- Positive value → emissions increased (worse performance).
- Negative value → emissions decreased (improved performance).
- Zero → no change compared to last year.

```
Emissions YoY Change =  
VAR PrevYear =  
    CALCULATE(  
        [Total Emission],  
        GHG_Emissions[Year] = SELECTEDVALUE(GHG_Emissions[Year]) - 1  
    )  
RETURN  
    [Total Emission] - PrevYear
```

```

Renewable YoY Change =
VAR PrevYear =
    CALCULATE(
        [Total Renewable],
        Renewable_Energy[Year] = SELECTEDVALUE(Renewable_Energy[Year]) - 1
    )
RETURN
    [Total Renewable] - PrevYear

```

Figure 35. Emission YOY Change & Renewable YOY Change

8. Total Emission & Total Renewable

This measure shows the total amount of greenhouse gas emissions & renew energy for the selected country, region, or year

```
Total Emission = SUM(GHG_Emissions[Value])
```

```
Total Renewable = SUM(Renewable_Energy[Value])
```

Figure 36. Total Emission & Total Renewable

9. Avg Emission By Country & Avg Emission By Country

This measure shows the average emission level for each country across all years.

```

Avg_Emission_By_Country =
CALCULATE (
    AVERAGE ( GHG_Emissions[Value] ),
    ALLEXCEPT ( GHG_Emissions, GHG_Emissions[Country] )
)

```

```

Avg_Renewable_By_Country =
CALCULATE (
    AVERAGE ( Renewable_Energy[Value] ),
    ALLEXCEPT ( Renewable_Energy, Renewable_Energy[Country] )
)

```

Figure 37. Avg Emission By Country & Avg Emission By Country

3.7 Define Relationships

Relationships with tables shown as follows before creating dashboard.

From Table	Column	To Table	Column	Type
GHG Emissions	Country Code	Dim Country	Country Code	Many-to-One, Single direction
Renewable Energy	Country Code	Dim Country	Country Code	Many-to-One, Single direction
GHG Emissions	Series Code	Dim Indicator	Series Code	Many-to-One
Renewable Energy	Series Code	Dim Indicator	Series Code	Many-to-One
GHG Emissions	Year	Dim Year	Year	Many-to-One
Renewable Energy	Year	Dim Year	Year	Many-to-One

Table 3. Define Relationships

4. Results and Visual Analysis

4.1 Dashboard Page 1: Global Greenhouse Gas Emissions Overview

This section outlines a comprehensive examination of the dashboard that includes a summary of all global greenhouse gas emissions, and how it reflects the patterns and trends of emissions over time, as shown in the data from 2000 to 2021 (i.e. cleaned and standardised) as well as through the eight different visuals focusing on specific aspects of the behavioural patterns associated with emissions. The various ways in which the dashboard depicts emissions behaviours, with regards to all emissions-producing countries, over time (i.e. how they compare their ranks based on their total amount of emissions), and how these countries have changed in terms of its total emissions from year to year, to become more concentrated geographically; provide an overview of how emissions have evolved over the years, what the major emitters are, and the periods and relative magnitude of the changes, to provide context for the evaluation of global efforts to mitigate climate change.

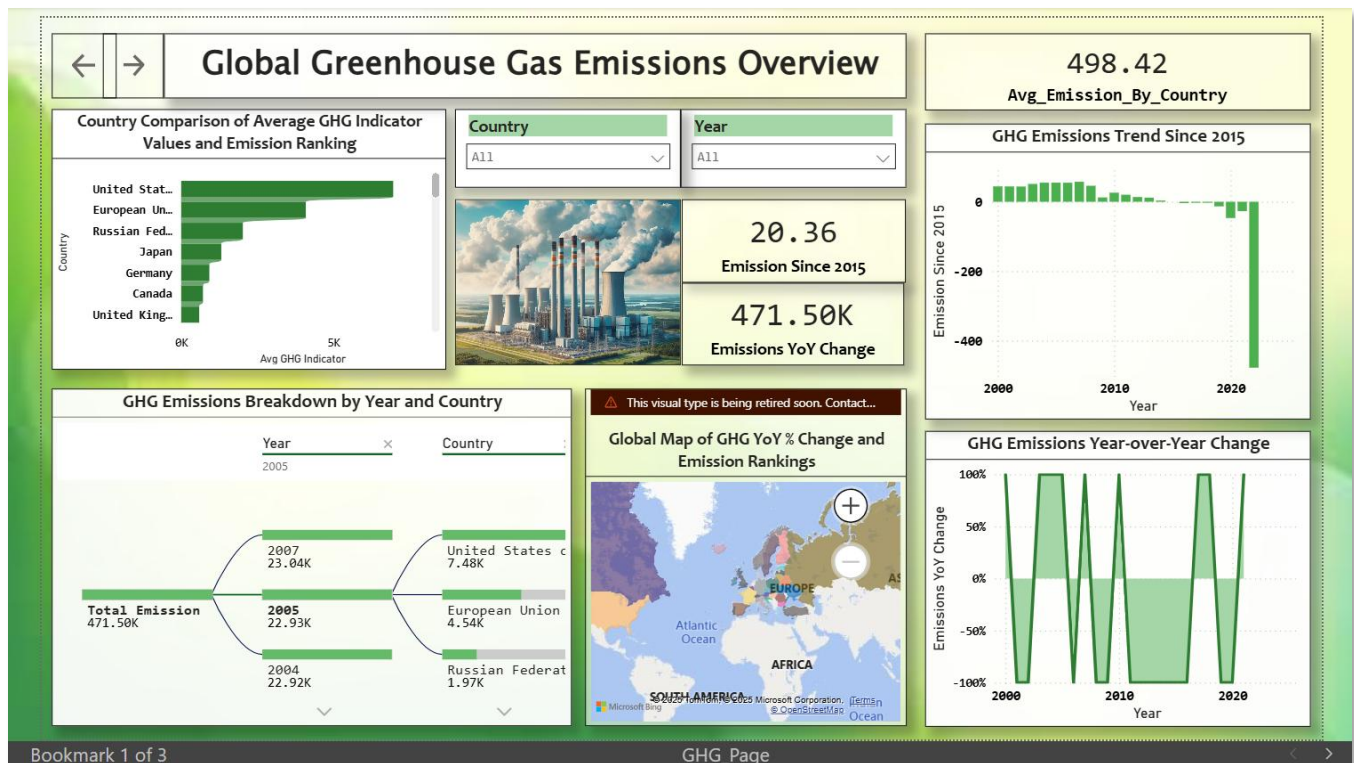


Figure 38. Dashboard Page 1: Global Greenhouse Gas Emissions Overview

The dashboard design presented in this page includes:

KPI Cards

- Average Emissions by Country
- Emissions Since 2015
- Year-over-Year Emissions Change

Bar Chart

- Country comparison of average GHG indicator values and emission rankings

Line / Column Chart

- Greenhouse gas emissions trend over time since 2015

Area / Line Chart

- Year-over-year change in greenhouse gas emissions

Decomposition Tree

- Breakdown of total emissions by year and country

Map Visualization

- Global geographic distribution of GHG year-over-year changes and country rankings

Slicers

- Country
- Year

1. KPI Card – Average GHG Emission by Country

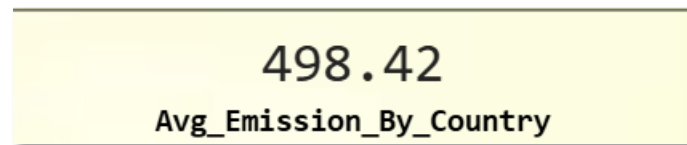


Figure 39. Average GHG Emission by Country

What the chart shows

- This KPI card displays the **average greenhouse gas emission value across all countries** based on the selected filters (country and year).

How the chart was created

- Go to Visualizations → select Card
- Add Avg_Emission_By_Country to the fields
- Keep all filters → ON

Interpretation

- The global average of 498.42 indicates that the intensity of emissions is still very high on a per-country basis; therefore, there is considerable disparity between the amount of emissions produced by each country versus the amount of renewable energy being utilised (the average of 27.51 for the example shown on the back of future dashboards). Therefore, this average is used as a benchmark in the interpretation of analyses at both the country and trend levels.

Insight

- The overall average of emissions is far above sustainable levels (even at the average level), so relying on incremental changes will not be enough.

2. Bar Chart – Country Comparison of Average GHG Indicator Values and Emission Ranking

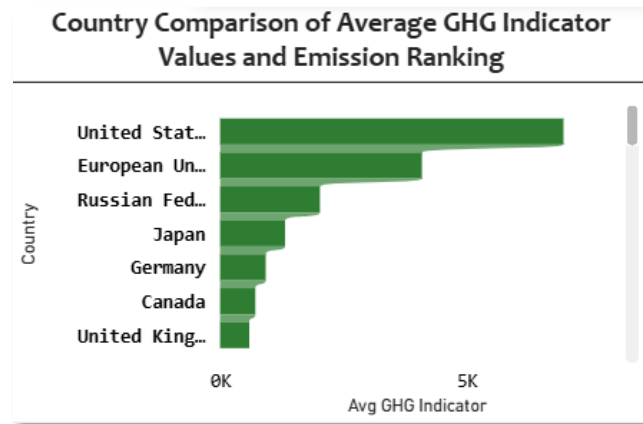


Figure 40. Country Comparison of Average GHG Indicator Values and Emission Ranking

What the visual shows

- This bar chart compares **average GHG emissions by country**, ranked from the highest to the lowest emitters.

How the visual was created (Steps)

- Choose **Clustered Bar Chart**.
- Add **Country** → Y-axis.
- Add **Avg GHG Indicator** → X-axis.
- Sort the chart in **descending order** by average emissions.
- Enable data labels for clarity.

Interpretation

- U.S.A. has the largest amount of emissions by far. They clearly dominate the graph.
- The E.U. and Russia are nearly as high as the U.S.A., demonstrating that they are also among the most significant emitters.
- Other countries, including Germany and Canada, as well as Great Britain, show a noticeable dip in emissions, suggesting a greater degree of efficiency, or perhaps more stringent emission limitations.
- The long drop from the leading emitters to the other countries indicates that a very small group is responsible for most of the emissions.

Key Insight

- The best approach to the reduction of global emissions is not to spread the responsibility evenly over every country but rather to focus on the few leading emitters.

3. Slicers – Country and Year

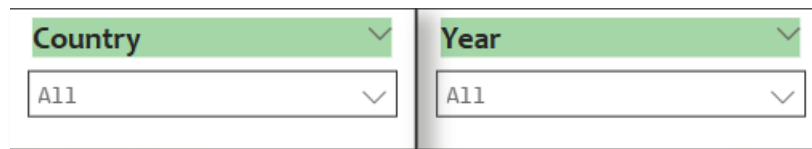


Figure 41. Slicers – Country and Year

What the visual shows

- These slicers allow users to filter the entire dashboard by **specific countries and years**.

How the visual was created (Steps)

- Choose **Slicer** visual.
- Add **Country** field to the slicer.
- Add **Year** field to a second slicer.
- Change slicer type to **Dropdown** for compact design.

Interpretation

- Slicers improve **interactivity and usability**, enabling focused analysis. For example, selecting a single country reveals how its emissions change over time and how it compares to global trends.

4. KPI Card – Emissions Since 2015 and Emission Year over Year Change

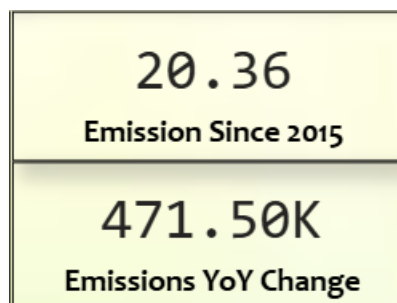


Figure 42. Emissions Since 2015 and Emission Year over Year Change

Emissions Since 2015

What the visual shows

- This KPI shows the **cumulative change in emissions since 2015**, the baseline year for the Sustainable Development Goals (SDGs).

How the visual was created (Steps)

- Choose **Card** visual.
- Add **Emission Since 2015** measure to *Values*.
- Format the card with a descriptive title and consistent colour scheme.

Interpretation

- A positive value (**20.36**) suggests that emissions have **increased since 2015**, indicating limited progress toward climate goals. This KPI is important for evaluating whether countries are moving in the right direction relative to global commitments.

Emission Year over Year Change

What the visual shows

- This card displays the **year-over-year change in emissions**, highlighting short-term fluctuations.

How the visual was created (Steps)

- Choose **Card** visual.
- Add **Emissions YoY Change** measure to *Values*.
- Apply conditional formatting to emphasize large changes.

Interpretation

- The high YoY change (**471.50K**) indicates **significant volatility** in emission levels. Such fluctuations may result from economic growth, industrial activity, or external shocks such as global crises.

5. Column Chart – GHG Emissions Trend Since 2015

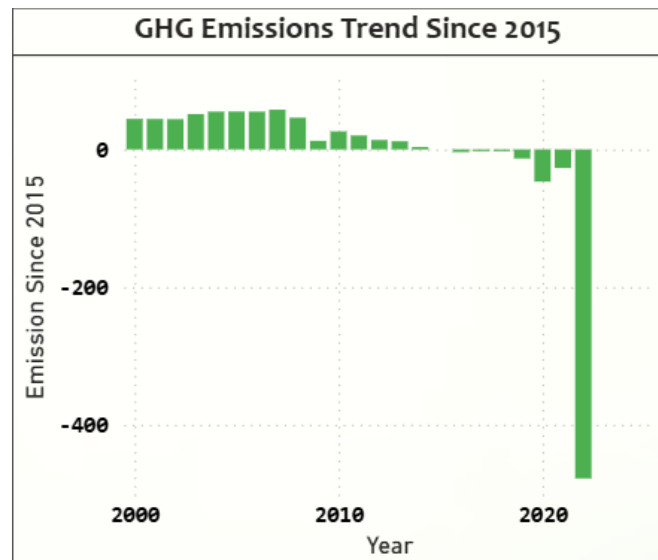


Figure 43. GHG Emissions Trend Since 2015

What the visual shows

- This chart visualises **annual emission changes from 2015 onward**.

How the visual was created (Steps)

- Choose **Column Chart**.
- Add **Year** → X-axis.
- Add **Emission Since 2015** → Y-axis.
- Enable gridlines and axis labels for readability.

Interpretation

- Post-2015, emissions begin to trend slowly upward indicating possibly either very little progress made at this stage or stagnation during the early years of SDGs.
- There is an apparent steep decline around 2020, which correlates strongly with the negative economic impacts resulting from COVID-19's shutdowns.
- Comparatively, this decline was far greater than any of the fluctuations observed on a year-to-year basis prior to this point; thus, indicating an external shock and not the success of a policy.

Key Insight

- The temporary drop in emissions due to external factors does not indicate progress towards long-term solutions for climate change.

6. Decomposition Tree – GHG Emissions Breakdown by Year and Country

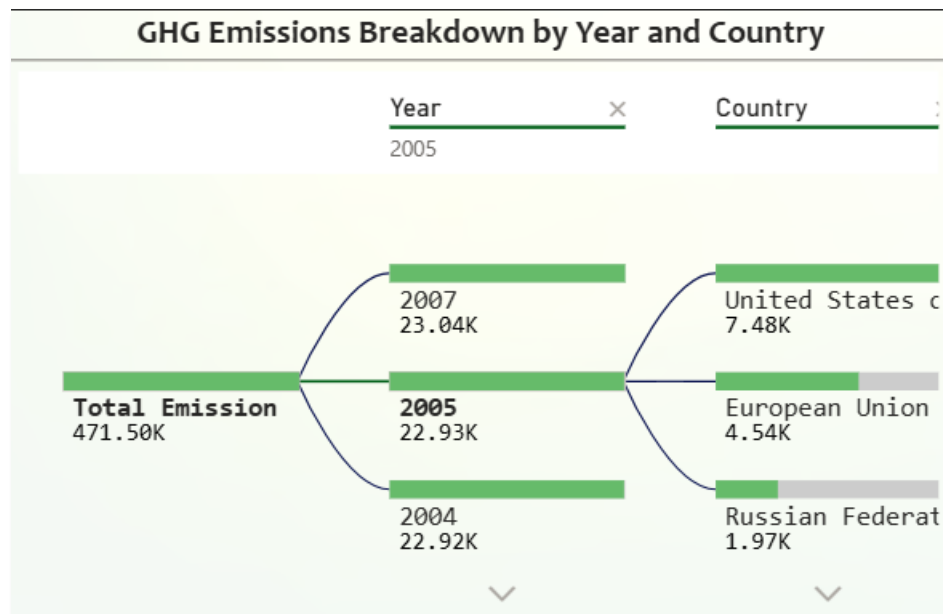


Figure 44. GHG Emissions Breakdown by Year and Country

What the visual shows

- These visual breaks down **total emissions into contributing years and countries.**

How the visual was created (Steps)

- Choose **Decomposition Tree**.
- Add **Total Emission** to *Analyze*.
- Add **Year** and **Country** as breakdown fields.
- Enable drill-down interaction.

Interpretation

- A handful of years (e.g., 2000 to 2004) has produced a disproportionate amount of total emissions.
- The United States and Europe are both frequent and significant sources of those emissions.

Key Insight

- International emissions reduction strategies must be tailored specifically for each country and for each year, rather than relying on a one-size-fits-all "cookie cutter" approach.

7. Map – Global Map of GHG YoY % Change and Emission Rankings

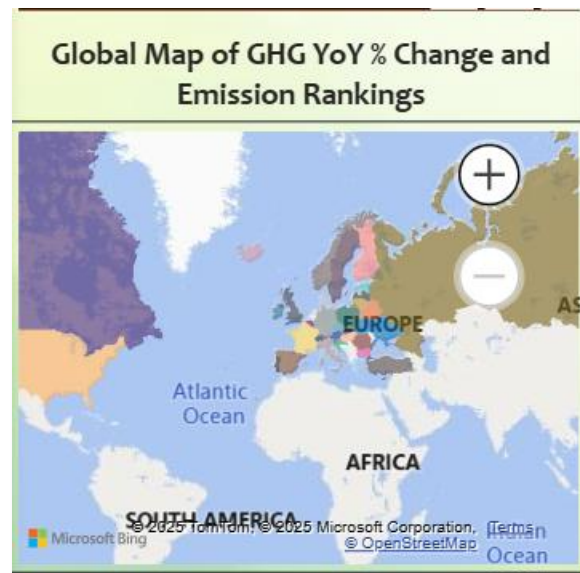


Figure 45. Global Map of GHG YoY % Change and Emission Rankings

What the visual shows

- This map illustrates the **geographic distribution of emissions and their year-over-year changes**.

How the visual was created (Steps)

- Choose **Map** visual.
- Add **Country** → Location.
- Add **Emissions YoY % Change** → Size/Color saturation.
- Enable zoom and tooltip details.

Interpretation

- The darker color of North America, Europe, and Asia indicates greater emissions and volatility of emissions.
- Africa has a lower intensity of emissions, which indicates that African countries have less industrial activity.

Key Insight

- Emission levels are concentrated in certain geographical areas, allowing for targeted policy development by region.

8. Area Chart – GHG Emissions Year-over-Year Change

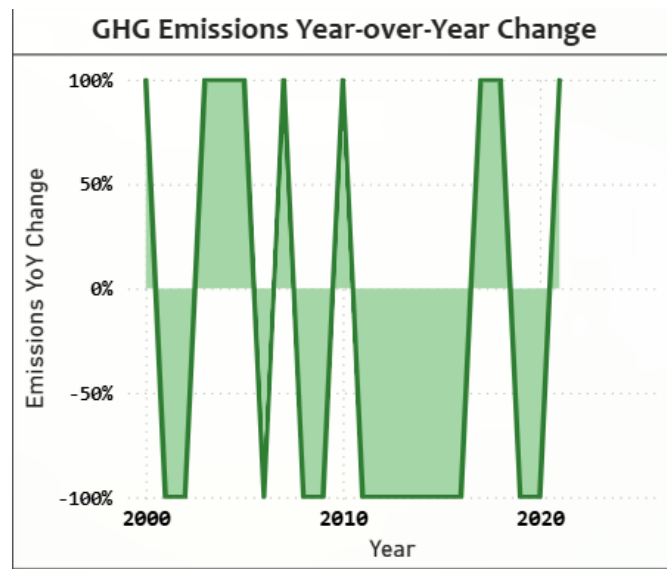


Figure 46. GHG Emissions Year-over-Year Change

What the visual shows

- This area chart shows the **percentage change in emissions from one year to the next**.

How the visual was created (Steps)

- Choose **Area Chart**.
- Add **Year** → X-axis.
- Add **Emissions YoY Change (%)** → Y-axis.
- Turn on **shaded area** in the Format pane.

Interpretation

- The graph exhibits extreme volatility with massive swings from highs of +100% to lows of -100%.
- There is no continuous decreasing trend; emissions tend to spike back up after they fall.
- Therefore, emission levels appear to be very economically sensitive with respect to the economic ups and downs.

Key Insight

- It is possible to conclude that emission reductions do not have structural characteristics and remain reliant upon economic situations.

Overall Interpretation

All the visuals combined provide a holistic overview of total global GHG emissions by integrating the key summary indicators, the ranking of emissions per country, historical and current trends and spatial distribution across the globe. The dashboard also indicates that, although there is evidence of reductions in global GHG emissions after the year 2015, the majority of the world's GHG emissions are still produced by a small number of large emitting nations, which indicates the persistent difficulty associated with achieving long-term goals related to sustainability.

4.2 Dashboard Page 2: Renewable Energy Progress Dashboard

This section covers the Renewable Energy Progress Dashboard, which provides an overview of how people around the world have embraced renewable energy and are progressing toward the set goals for a sustainable energy future (using data for the period from 2000 to 2021). The Renewable Energy Progress Dashboard provides information in several ways to help you see how different countries are performing when it comes to renewable energy, how they have grown since the Sustainable Development Goal (SDG) baseline year of 2015, how their progress changes from one year to the next, and how they are developing in different parts of the world. All this information is provided in graphical format, making it easy for you to compare the relative levels of renewable energy adoption in various countries at any given time, as well as to assess whether global efforts are keeping pace with what would be needed in order to meet long-term emissions reduction targets.

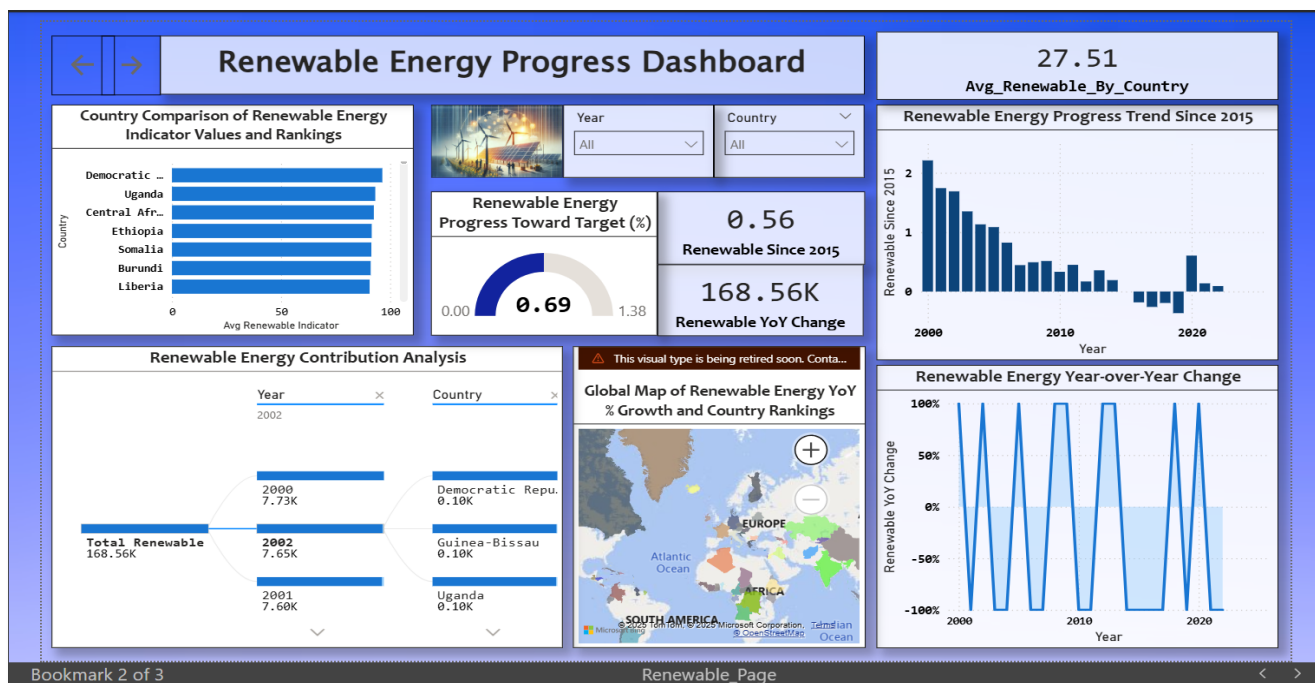


Figure 47. Dashboard Page 2: Renewable Energy Progress Dashboard

The dashboard design developed for this page includes:

KPI Cards

- Average Renewable Energy by Country
- Renewable Energy Since 2015
- Year-over-Year Renewable Energy Change

Bar Chart

- Country comparison of renewable energy indicator values and rankings

Gauge Chart

- Renewable energy progress toward target (%)

Column Chart

- Renewable energy progress trend since 2015

Line / Area Chart

- Year-over-year change in renewable energy indicators

Decomposition Tree

- Renewable energy contribution analysis by year and country

Map Visualization

- Global distribution of renewable energy growth and country rankings

Slicers

- Country
- Year

1. KPI Card – Average Renewable Energy Indicator by Country

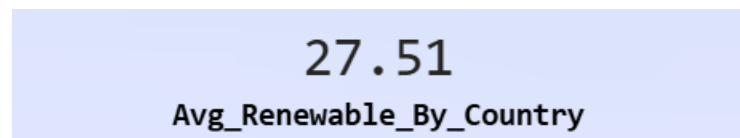


Figure 48. Average Renewable Energy Indicator by Country

What the visual shows

- This KPI card displays the **average renewable energy indicator value across all countries**, based on the applied year and country filters.

How the visual was created (Steps)

- Choose **Card** from the Visualizations pane.
- Drag **Avg_Renewable_By_Country** into *Values*.
- Apply **Year** and **Country** slicers for interactivity.
- Format the card with a large font and descriptive title.

Interpretation

- The average value of 27.51 signifies that globally, less than one-third of the total energy used is supplied by renewable sources.
- On the dashboard (498.42), the contrast between renewable energy adoption and greenhouse gas emissions is clear. The immensity of the two values demonstrates that there is a substantial amount of clean and potentially clean energy being produced around the globe but not enough to offset emissions.
- The low average value of the above supports the conclusion that most areas still utilize predominantly fossil fuels for most of their energy needs; they continue to rely on renewable energy as a supplementary form of energy rather than as a primary energy source.

Key Insight

- The current amount of renewable energy produced worldwide is not enough reduction to bring worldwide emissions back down to acceptable levels, requiring the rapid increase of renewable energy production in addition to the continued reduction of other forms of energy.

2. Bar Chart – Country Comparison of Renewable Energy Indicator Values and Rankings

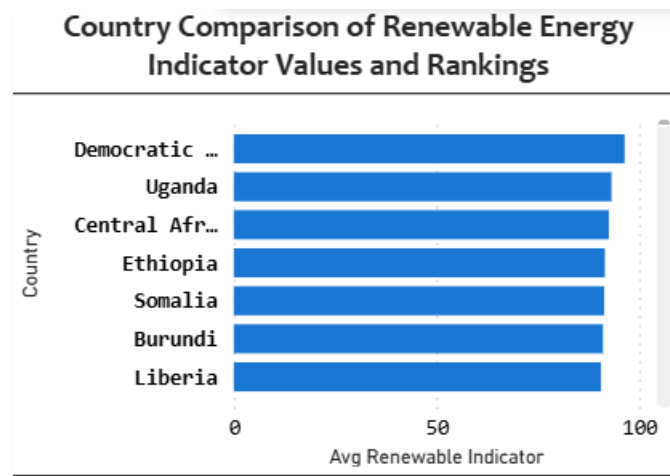


Figure 49. Country Comparison of Renewable Energy Indicator Values and Rankings

What the visual shows

- This bar chart ranks countries based on their **average renewable energy indicator values**, highlighting relative performance.

How the visual was created (Steps)

- Choose **Clustered Bar Chart**.
- Add **Country** → Y-axis.
- Add **Avg Renewable Indicator** → X-axis.
- Sort the chart in **descending order**.
- Enable data labels for clarity.

Interpretation

- Countries including Uganda, Ethiopia, and Democratic Republic of Congo are among the highest ranking nations in reliance on renewable energy sources. This indicates some developing countries are leading in the adoption of renewable energy, primarily as a result of their use of hydroelectric and biomass rather than dependent on fossil fuels.

3. Slicers – Year and Country

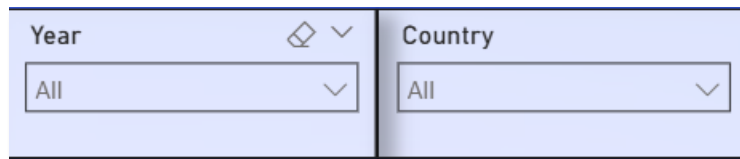


Figure 50. Slicers – Year and Country

What the visual shows

- These slicers allow users to dynamically filter renewable energy data by **specific years and countries**.

How the visual was created (Steps)

- Choose **Slicer** visual.
- Add **Year** to one slicer and **Country** to another.
- Change slicer style to **Dropdown** for compact layout.

Interpretation

- Slicers enhance **user-driven exploration**, enabling focused analysis such as examining renewable growth trends for a single country or comparing different time periods.

4. Gauge Chart – Renewable Energy Progress Toward Target (%)

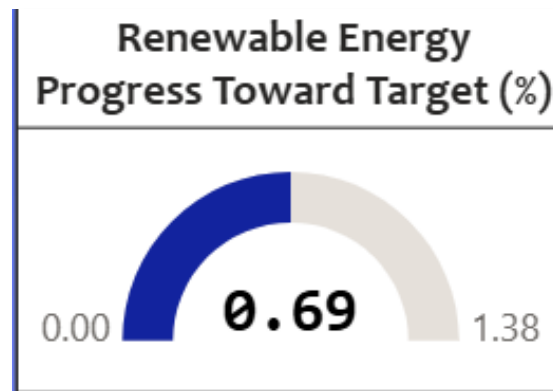


Figure 51. Renewable Energy Progress Toward Target (%)

What the visual shows

- This gauge visual represents **progress toward a predefined renewable energy target**, expressed as a percentage.

How the visual was created (Steps)

- Choose **Gauge** visual.
- Add **Progress Toward Target (%)** → Value.
- Set **minimum**, **maximum**, and **target** values in the Format pane.
- Adjust colours to highlight progress zones.

Interpretation

- The gauge value (**0.69**) indicates **partial progress toward renewable energy targets**. While improvements are evident, the indicator shows that global renewable adoption remains below desired sustainability thresholds.

5. KPI Card – Renewable Energy Since 2015 and Renewable year over Year Change

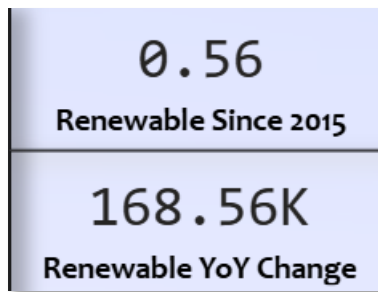


Figure 52. Renewable Energy Since 2015 and Renewable year over Year Change

Renewable Energy Since 2015

What the visual shows

- This card shows the **cumulative change in renewable energy adoption since 2015**, the SDG baseline year.

How the visual was created (Steps)

- Choose **Card** visual.
- Add **Renewable Since 2015** measure.
- Format the card with consistent theme colours.

Interpretation

- The value (**0.56**) indicates modest progress since 2015, suggesting that while renewable energy adoption is increasing, the pace may be insufficient to meet long-term climate goals.

Renewable year over Year Change

What the visual shows

- This KPI highlights the **annual change in renewable energy indicators**.

How the visual was created (Steps)

- Choose **Card** visual.
- Add **Renewable YoY Change** measure.
- Apply conditional formatting to emphasise growth or decline.

Interpretation

- A value of **168.56K** reflects **significant year-to-year variation**, indicating that renewable expansion is uneven and influenced by economic conditions, infrastructure investment, and policy changes.

6. Column Chart – Renewable Energy Progress Trend Since 2015

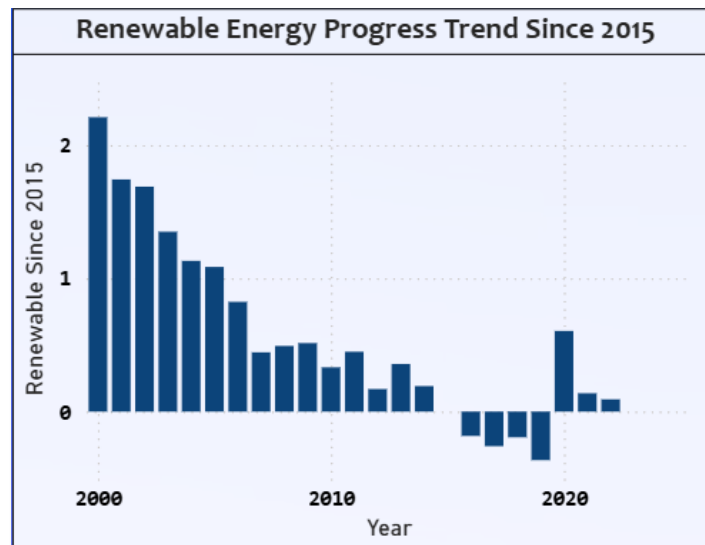


Figure 53. Renewable Energy Progress Trend Since 2015

What the visual shows

- This column chart displays **renewable energy trends from 2000 onwards**, with emphasis on changes after 2015.

How the visual was created (Steps)

- Choose **Column Chart**.
- Add **Year** → X-axis.
- Add **Renewable Since 2015** → Y-axis.
- Enable gridlines and axis titles.

Interpretation

- The bars represent a positive trend in growth from 2015 forward, implying that policies are consistent with SDG 13.
- In addition, the increase is not ample nor consistent; therefore, it is incremental but at times, it is marginally improved or decreased.
- Since there are no sudden upward trends in growth, this supports that the rate of infrastructure development is very slow.

Key Takeaway

- The improvement towards renewable energy since 2015 has been a continuous increase with little change, rather than groundbreaking changes.

7. Decomposition Tree – Renewable Energy Contribution Analysis

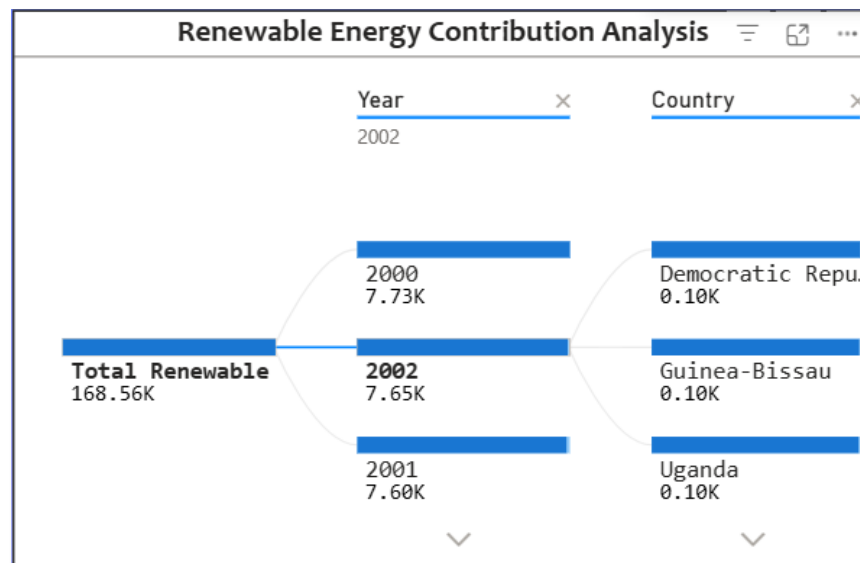


Figure 54. Renewable Energy Contribution Analysis

What the visual shows

- These visual breaks down **total renewable energy contributions** by **year and country**.

How the visual was created (Steps)

- Choose **Decomposition Tree**.
- Add **Total Renewable** → Analyze.
- Add **Year** and **Country** as breakdown dimensions.
- Enable drill-down interaction.

Interpretation

- Relatively few countries provide a great deal of total renewable energy.
- Many other countries provide relatively little renewable energy, demonstrating the global disparity in their use of renewable energy.

Key Insights

- Renewable energy growth is occurring mostly in only a small number of countries, which diminishes its overall impact.

8. Map – Global Map of Renewable Energy YoY % Growth and Rankings



Figure 55. Global Map of Renewable Energy YoY % Growth and Rankings

What the visual shows

- This map visualises the **geographic distribution of renewable energy growth rates**.

How the visual was created (Steps)

- Choose **Map** visual.
- Add **Country** → Location.
- Add **Renewable YoY % Growth** → Size or Color saturation.
- Enable zoom and tooltips.

Interpretation

- The growth of hydropower has contributed to strong economic development in Africa and parts of Asia.
- High hydropower expansion is one of the main contributors to strong growth around Africa & Asia.
- Europe & North America are experiencing moderate or stable growth due to their respective market saturation.

Insights

- As developing countries continue to invest in their own growth, it is expected that future growth opportunities for renewables will come from developing areas rather than traditional industrial countries.

9. Line Chart – Renewable Energy Year-over-Year Change (%)

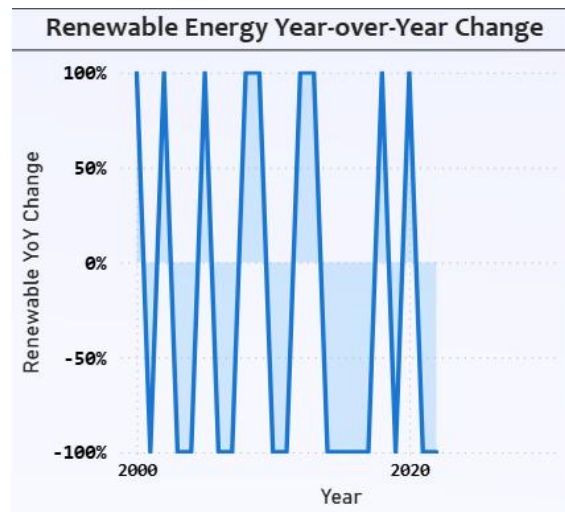


Figure 56. Renewable Energy Year-over-Year Change (%)

What the visual shows

- This line chart illustrates **percentage changes in renewable energy adoption over time.**

How the visual was created (Steps)

- Choose **Line Chart**.
- Add **Year** → X-axis.
- Add **Renewable YoY Change (%)** → Y-axis.
- Turn on markers and smooth lines in Format.

Interpretation

- The fluctuations depicted in the graph illustrate why renewable energy isn't usually developed on a steady upward-sloping trajectory.
- Instead, large-scale development of renewable energy will result in dramatic increases in growth until funding or political conditions change.
- They illustrate the types of events that lead to rapid escalation in the deployment of solar and wind technologies, and what happens next when the environment for deploying renewables is not favorable.

Key Insights

- Renewable energy has been growing rapidly: dramatically increasing the rate of development of renewable energy. Until now, rapid growth has not resulted in a steady and/or ongoing growth (i.e., increased investment). Rapid growth is susceptible to various types of disruptions (e.g., political, financial).

Overall Interpretation

The global renewable energy tracker dashboard gives you all the information you need to understand how well or poorly the world is adopting renewable energy sources. The dashboard displays data in several different ways — through KPI summaries, country ranks, historical trends and geographical locations. The data show that even though global renewable energy generation is on the rise, adoption levels are inconsistent, showing that more coordinated efforts and continued investment in policies are necessary.

9.1 Dashboard Page 3: Comparative Analysis & Deep Dive Explorer

This part of the report is an overview of the analysis phase of the Comparative Analysis (CA) and Deep Dive Explorer (DDE). It combines GHG emission indicators with renewable energy indicators to allow for comparisons and relationships between different years/countries. The visual representations included in the CA and the DDE provide a way to compare total and average values of both types of indicators, see how they changed year on year, identify international trends in emissions and renewable energy development, see where emissions and renewable energy occur in relation to each other (through geographic distribution), and show the relationship(s) between greenhouse gas emissions and the adoption of renewable energy. The combination of these visual representations allows users to see how renewable energy growth correlates with emission outcomes as well as finding examples of both significant and unexpected emissions/renewable energy combinations that are important for evaluating climate policy through an evidence-based approach.

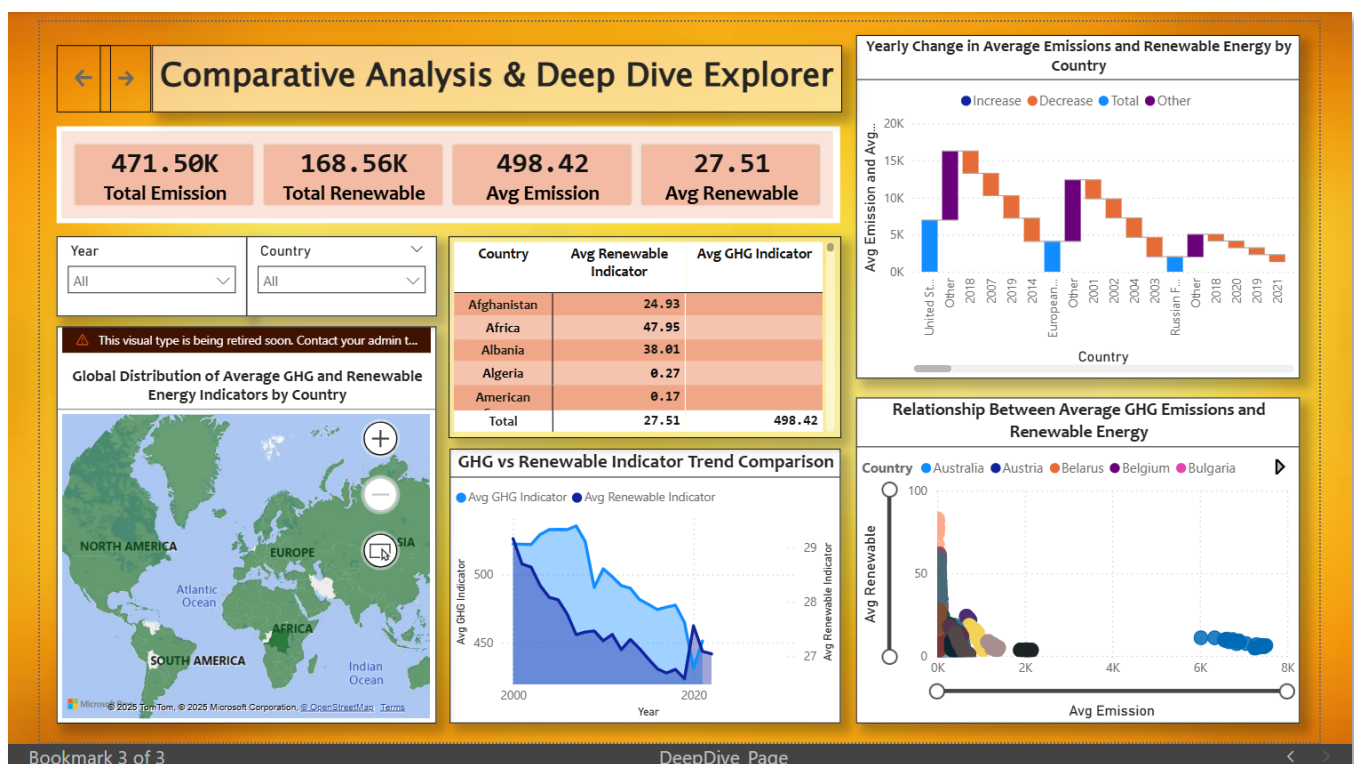


Figure 57. Dashboard Page 3: Comparative Analysis & Deep Dive Explorer

The dashboard design for this page includes:

KPI Cards

- Total Emissions
- Total Renewable Energy
- Average Emissions
- Average Renewable Energy

Table

- Country-level comparison of average GHG emission indicators and average renewable energy indicators

Waterfall Chart

- Yearly change in average emissions and renewable energy by country

Line / Area Chart

- Trend comparison between GHG emissions and renewable energy indicators over time

Scatter Plot

- Relationship between average GHG emissions and renewable energy adoption across countries

Map Visualization

- Global distribution of average emissions and renewable energy indicators

Slicers

- Country
- Year

1. KPI Cards – Overall Emissions and Renewable Energy Summary

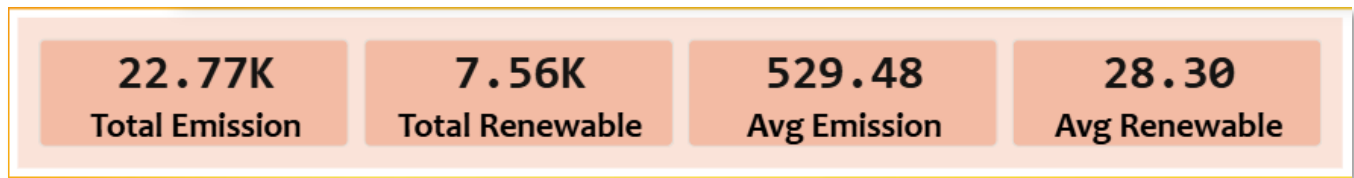


Figure 58. Overall Emissions and Renewable Energy Summary

What the visuals show

- These four KPI cards summarise **key global metrics**, providing a snapshot comparison of emissions and renewable energy indicators.
- **Total Emission:** 471.50K
- **Total Renewable:** 168.56K
- **Average Emission:** 498.42
- **Average Renewable:** 27.51

How the visuals were created (Steps)

- Choose **Card** visual.
- Add **Total Emission**, **Total Renewable**, **Avg Emission**, and **Avg Renewable** measures separately.
- Apply **Year** and **Country** slicers to affect all cards.
- Format cards with consistent background colour and large font size.

Interpretation

- Between total emissions and total renewable energy generation are nearly a factor of three separating them from each other.
- Average emissions, on the other hand, are greater than renewable average emissions of over eighteen times average emissions leading to structural imbalance.
- The KPI contour plot provides a visual representation of the magnitude of the disparity between emissions and mitigation activities.

Key Insight

- The growth of renewable energy sources is not directly proportional to the amount of emissions being generated making them less effective for climate change mitigation.

2. Slicers – Year and Country Filters



Figure 59. Slicers – Year and Country Filters

What the visuals show

- The slicers enable users to filter the dashboard by **specific years or countries**, supporting deep, focused analysis.

How the visuals were created (Steps)

- Choose **Slicer** visual.
- Add **Year** to one slicer and **Country** to another.
- Set slicer type to **Dropdown** for compact layout.
- Enable *Sync slicers* across pages.

Interpretation

- Slicers enhance **interactivity and flexibility**, allowing users to isolate trends for individual countries or time periods and compare performance across multiple dashboard pages.

3. Table – Country-Level Average Emission and Renewable Indicators

Country	Avg Renewable Indicator	Avg GHG Indicator
Afghanistan	36.66	
Africa	50.70	
Albania	33.67	
Algeria	0.47	
American	0.00	
Total	28.30	529.48

Figure 60. Country-Level Average Emission and Renewable Indicators

What the visual shows

- This table presents **country-wise average renewable energy indicators alongside average GHG emission indicators**.

How the visual was created (Steps)

- Choose **Table** visual.
- Add **Country**, **Avg Renewable Indicator**, and **Avg GHG Indicator** to Values.
- Apply conditional formatting to highlight variations.
- Enable totals at the bottom.

Interpretation

- The majority of countries' electricity supply comes from fossil fuels.
- A few countries hold a good balance of renewable energy compared to their total electricity usage.
- The way the table has been set up makes it easy to see and compare numbers. Even if a country ranks higher in terms of percentage of renewable energy used, it still emits a lot of greenhouse gases.

Trends in Observations

- There are no clear clusters or groups of countries that show a continuity of low emissions/high number of renewables in use.
- Countries that have a high number of emissions may not necessarily have a high number of renewables available to them.

Key Takeaway

- There continues to be a lack of alignment in the transition to a new energy economy across the globe, and no one model represents the "best practices" for all countries.

4. Waterfall Chart – Yearly Change in Average Emissions and Renewable Energy by Country

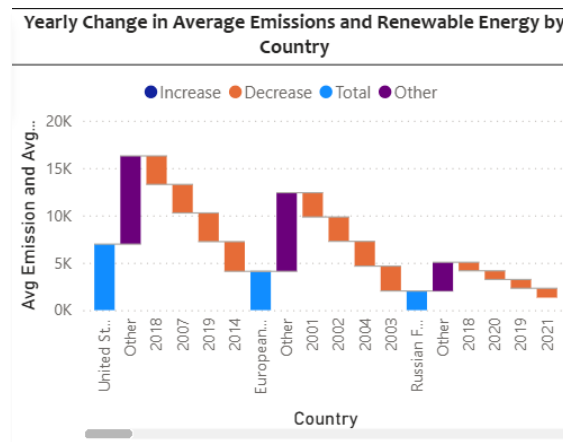


Figure 61. Yearly Change in Average Emissions and Renewable Energy by Country

What the visual shows

- This waterfall chart illustrates **year-by-year increases and decreases** in average emissions and renewable energy indicators across selected countries.

How the visual was created (Steps)

- Choose **Waterfall Chart**.
- Add **Country / Year** → X-axis.
- Add **Avg Emission and Avg Renewable Change** → Y-axis.
- Set categories for **Increase, Decrease, and Total**.
- Enable data labels and legend.

Interpretation

- The alternation between positive and negative values in the waterfall chart indicates that emissions reductions are not steady or predictable, and they exhibit a high degree of variability.
- In a year when emissions were reduced, the majority of those reductions were likely offset by increased emissions the following year.
- The improvement in emissions appears smaller and less predictable than the increase in renewable energy generation.

General Observations

- Emission reductions have not been sustained, and improvements have been sporadic and often reversed.

Main Takeaway(s)

- Climate progress continues to be unstable and reversible, subject to a variety of external influences.

5. Map – Global Distribution of Average GHG and Renewable Energy Indicators

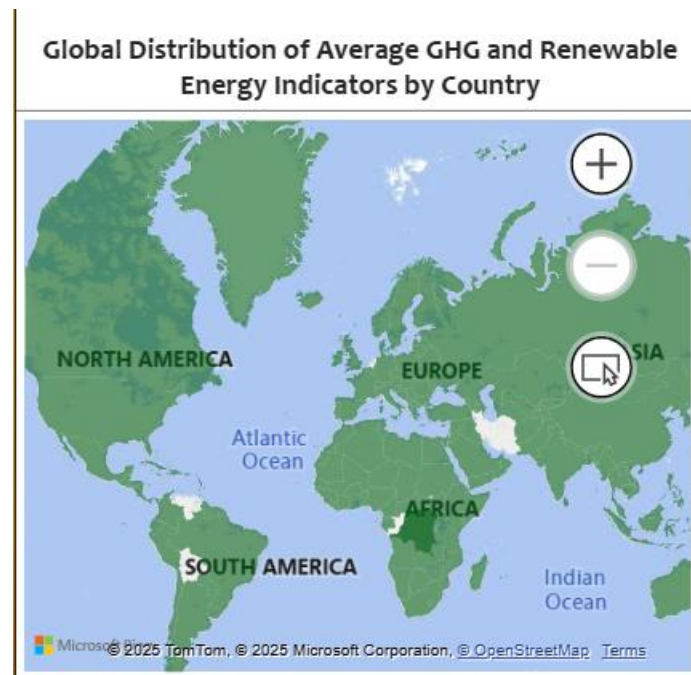


Figure 62. Global Distribution of Average GHG and Renewable Energy Indicators

What the visual shows

- This map visualises the **geographical distribution of average emissions and renewable indicators by country**.

How the visual was created (Steps)

- Choose **Map** visual.
- Add **Country** → Location.
- Add **Avg GHG Indicator** and **Avg Renewable Indicator** → Tooltip or Size/Color saturation.
- Enable zoom controls and map labels.

Interpretation

- The map reveals **regional patterns**, with higher emission concentrations in industrialised regions and comparatively higher renewable indicators in parts of Africa and South America. This spatial contrast highlights **structural differences in energy systems and development stages**.

6. Line Chart – GHG vs Renewable Indicator Trend Comparison

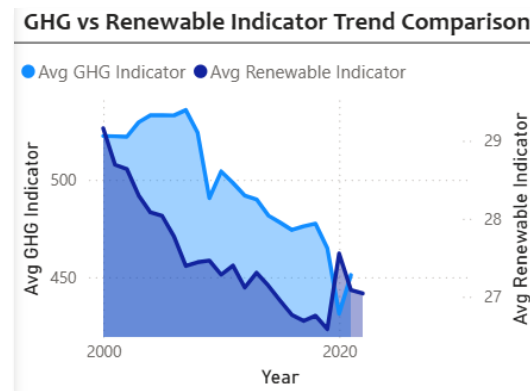


Figure 63. GHG vs Renewable Indicator Trend Comparison

What the visual shows

- This line chart compares **long-term trends** of average GHG emissions and renewable energy indicators over time.

How the visual was created (Steps)

- Choose **Line Chart**.
- Add **Year** → X-axis.
- Add **Avg GHG Indicator** and **Avg Renewable Indicator** → Y-axis.
- Turn on **Shaded area** in the Format pane.

Interpretation

- Incremental reductions in emissions remain significantly greater than any earlier renewable energy levels across all years.
- Incremental growth in renewable energy levels are erratic but overall very slow and therefore "gently" slope upward compared to the "steep" slope up from all emissions, making it clear visually that the two substances do not even begin to meet, indicating little if anything has been accomplished in decreasing emissions through any alternative or "cleaner" forms of energy.

General Observations

- Emission reductions occur incrementally and generally fairly slowly
- The rate at which renewable energy can be increased does not accelerate fast enough to offset total emissions

Key Takeaway

- If the rate of growth in renewable energy continues to be only incremental (or even slower), there will always be an ongoing gap between emissions and renewable energy.

7. Scatter Plot – Relationship Between Average GHG Emissions and Renewable Energy

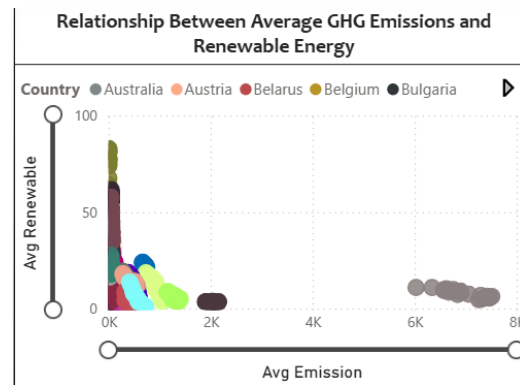


Figure 64. Relationship Between Average GHG Emissions and Renewable Energy

What the visual shows

- This scatter plot examines the **relationship between average emissions and renewable energy adoption across countries**.

How the visual was created (Steps)

- Choose **Scatter Chart**.
- Add **Avg Emission** → X-axis.
- Add **Avg Renewable** → Y-axis.
- Add **Country** → Legend.
- Enable zoom sliders for both axes.

Interpretation

- Countries predominantly fall into the low-renewable/high emissions quadrant of the matrix.
- Few countries are located in the high-renewable/low emissions quadrant, showing that countries with optimal energy profiles are few and far between.
- No strong inverse correlation exists between renewable adoption and reduced emissions, which implies that an increase in renewable energy adoption does not necessarily equal decreased emissions.

Patterns Observed

- While an increase in the use of renewable energy allows for emission reductions, it is not the only condition for achieving such results.
- Structural factors, such as high consumption levels and industrial intensity, are generally the dominant drivers of emissions.

Key Insight

- Effective climate strategies must take an integrated approach to renewable energy expansion, energy efficiency, and demand reduction.

10. Discussion

10.1 Global Climate Indicators Overview

An Integrated Overview of Global Climate Indicators combines information of Global Greenhouse Gas (GHG) emissions and the extent to which countries have integrated renewable energy into their economies from 2000 through 2021 to establish long term environmental trends. This analysis presents three dashboards showing GHG Emission Intensity, Growth of Renewable Energy and Comparison of Country or Regional Integration.

10.2 GHG Emission Intensity and Distribution

The first dashboard depicts that ,even though most of the world's GHG emissions come from a limited number of countries, there is no long-term decrease in the total amount emitted by these countries because the average GHG Emission Indicator Value for the world's highest emitting countries is 498.42 gCO₂ per capita with total emissions of 471.50K gCO₂; this indicator indicates that, although emissions have decreased slightly since 1990 for each country respectively, the decrease in emissions has not been consistent across all regions or time frames. Thus, emissions continue to be reduced with some countries and regions experiencing greater emissions reductions than others.

10.3 Renewable Energy Use Trends

The second dashboard demonstrates that there has been overall gradual growth in the amount of renewable energy used as represented by the indicator value of 27.51 gCO₂ (which represents a total of 168.56K gCO₂ in total contribution of renewable energy) over the past 20 years; however, this growth has not occurred uniformly across all countries and the degree to which renewable energy has been adopted varies greatly from country to country. For example, while there are a small number of countries with very high renewable energy use and very low levels of GHG emissions, most countries have not yet reached these levels of adoption.

10.4 Comparing Countries' Performance

Dashboard three demonstrates that the countries demonstrating the highest rates of growth in renewable energy adoption do not necessarily correlate with countries displaying the lowest total GHG emissions, as several countries demonstrating high levels of renewable energy usage still have substantial levels of total GHG emissions and therefore do not attain proportional emission reductions as a result of advancing toward renewable energy deployment alone.

10.5 General Observation

Across all three dashboards, it is clear that there are large differences between countries in their approach to environmental performance and progress. The analysis of emissions reductions worldwide also highlights that countries are transitioning from fossil fuel-based energy systems to sustainable energy systems at very different speeds.

11. Recommendation

11.1 Increase the rate of expansion of renewable energy sources.

Countries with high and continuing greenhouse gas emissions should be focusing on increasing their renewable energy capacity beyond what currently exists. The dashboard data demonstrates that the current renewable energy use is not at a sufficient level to provide a counterbalance to the increase in greenhouse gas emissions.

11.2 Focus strategies targeting higher emitting countries.

Due to a small number of countries producing many global greenhouse gas emissions, intervention strategies in these countries may produce greater global reductions than generic approaches to policy.

11.3 Combine Renewable Energy Adoption with Energy Efficiency Improvements.

When using renewable energy, it is also necessary to improve energy efficiency and demand management of the entire energy system. The data in the dashboard demonstrates that renewable energy only does not always result in reduced emissions.

11.4 Strengthen the Monitoring of Greenhouse Gas Emissions through Data Dashboards.

The use of interactive data dashboards, like those developed through this study, will allow for the continuous monitoring of both greenhouse gas emissions and renewable energy trends and make timely evaluations of progress and identify barriers to success.

11.5 Share Best Practices Across Countries.

Lower emitting and higher renewable energy-using countries can be valuable benchmarks for other countries. The exchange of information on "best practices" can help to accelerate the rate of improvement in countries that lag.

12. Limitations

12.1 Country-level Aggregate Data

The analysis uses country-level data; therefore, regional or sectoral variations in emissions and renewable energy adoption may be overlooked.

12.2 Scope of Indicators is Limited

Only those indicators relating to greenhouse gas emissions and renewable energy were included, and related areas such as energy efficiency, economic structure, population growth, and technological advancement were not considered.

12.3 Variability Across Countries in Data Collection

The variability in methods of gathering and reporting data in addition to the frequency of reporting among countries creates disparities in the consistency and comparability of indicators across time.

12.4 No Technology-Specific Breakdown for Renewable Energy Data

The renewable energy data is not broken down by technology; therefore, it is difficult to evaluate the effectiveness of individual technologies for emission reductions.

12.5 Correlation Not Causation

The dash boards present correlations and patterns related to emissions and renewable energy, but there are no causal relationships identified between these two variables.

13. Conclusion

Through the use of SQL for data preparation and interactive dashboards created in Power BI, this study employs a data-driven approach to examine global greenhouse gas emissions and the growth of renewable energy from 2000-2021.

The analysis integrates datasets on both greenhouse gas emissions and renewable energy usage to give a complete and accurate account of long-term trends, differences between countries, and the relative environmental performance of countries.

While the research indicates that global greenhouse gas emissions are concentrated in a small number of countries, the rate of adoption of renewable energy sources has not kept pace with the number of countries making significant efforts to utilize them. While many countries have made progress toward adopting renewable energy, their adoption rates do not always correlate to a proportional reduction in greenhouse gas emissions. Furthermore, this comparative analysis demonstrates that the relationship between renewable energy adoption and greenhouse gas emissions is complicated by a variety of factors, including energy consumption, industrial production and structures.

Overall, the dashboards turn complex global data sets into visual insight that is important for informing analysis of climate-related indicators. Data-driven research is beneficial in understanding global sustainability issues and serves as an appropriate foundation for continued research and action toward establishing sustainable policies and practices.